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DOES GENERATIVE AI NARROW EDUCATION-BASED PRODUCTIVITY GAPS? EVIDENCE FROM A RANDOMIZED EXPERIMENT

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Does Generative AI Narrow Education-Based Productivity Gaps? Evidence from a Randomized Experiment

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ABSTRACT

Does generative artificial intelligence (AI) reinforce or reduce productivity differences across workers?Existing evidence largely studies AI within firms and occupations, where organizational selection compresses educational heterogeneity, leaving unclear whether AI narrows productivity gaps across individuals with different levels of education. A related question is whether the productivity gains from AI reflect productive use of the tool or mere delegation, fading once AI is unavailable, and whether the answer differs across education groups. We address both questions using a randomized online experiment outside firms, in which 1,174 adults aged 25–45 complete an incentivized, workplace-style business problem-solving task with or without a generative-AI assistant, followed by a non-AI-assisted follow-up module. AI increases performance for all participants, with substantially larger gains for lower-education individuals. In the control group without AI, higher-education participants outperform lower-education participants by 0.548 standard deviations; with AI, this gap falls to 0.139 standard deviations, closing three-quarters of the initial gap. Chat-log measures help interpret both why this gap narrows and why it does not disappear: lower-education participants successfully obtain substantial assistance from AI, while higher-education participants use the tool more effectively across several margins. The follow-up results show that these gains are not purely driven by delegation; treated participants do not perform worse than controls once AI is removed, and lower-education participants retain part of their gain, although a sizable education gap re-emerges. Consistent with this interpretation, intensive use of the assistant produces strong task performance regardless of how much time and effort participants devote to the task, but follow-up performance is substantially higher only when intensive AI use is combined with sustained effort on the task. We interpret these findings as evidence that generative AI narrows effective productivity differences in task execution while underlying human-capital differences continue to shape unassisted performance and effective use of the tool.

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A randomized controlled trials registry entry is available at
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1 Introduction

Technological change has long been a driver of inequality in labor markets. Historically, innovations have often disproportionately benefited highly educated workers, contributing to widening gaps in employment and earnings (Machin and Van Reenen, 1998; Goldin and Katz, 2008; Acemoglu and Autor, 2011). Generative artificial intelligence (AI) now presents a new paradigm: unlike many previous technologies, these systems can perform complex cognitive tasks. This raises the possibility that generative AI could be "skill-democratizing" rather than skill-biased, as suggested by Autor (2024), enabling workers with limited formal training to perform tasks previously requiring extensive education and thus empowering workers across the education distribution. Acemoglu (2025), by contrast, cautions that AI's effects on labor markets may be more modest and unequally distributed. Whether AI weakens the link between formal education and productivity or instead reproduces existing skill advantages in a new form remains an open empirical question. Direct experimental evidence on how AI affects productivity differences across education groups, and through what mechanisms, remains scarce.

This paper asks a simple but consequential question: does generative AI reduce or reinforce productivity gaps across individuals with different levels of formal education, and how? We also ask whether the productivity gains from AI reflect productive use of the tool or merely delegation to AI, fading once AI is unavailable. Existing evidence largely comes from workplace settings and specific occupations, where hiring and task assignment compress educational heterogeneity. These studies provide valuable evidence on variation within relatively homogeneous groups, but by design are less well suited to address whether AI narrows or widens the education-based productivity gaps that feature prominently in the inequality debate (Goldin and Katz, 2008). To address these questions, we designed a randomized online experiment outside firms in which high-school-educated and postsecondary-educated adults complete the same incentivized business problem-solving task either with or without access to a GPT-based assistant. Holding the task fixed and randomizing access across education-defined groups identifies how AI changes the productivity advantage associated with formal education, namely the between-group performance gap. We pair the task with an immediate non-AI-assisted follow-up module to distinguish between gains that are solely contemporaneous and those that partly carry over once the tool is removed. The design deliberately trades some organizational realism for greater educational heterogeneity, which is typically compressed within firms.[1]

Our preregistered experiment includes 1,174 individuals aged 25 to 45 from Argentina, divided into low- and high-education groups. We define low-education

[1]Experimental designs differ from naturally occurring settings along multiple dimensions (Harrison and List, 2004); ours relaxes several of these to preserve greater educational variation.

individuals as those who have only a high-school diploma or have completed less than half of a postsecondary degree, and high-education individuals as those who have completed at least half of such a program.[2] Participants solve a self-contained workplace-style problem that requires reading text, a figure, and a table, diagnosing the problem, and drafting an email that proposes a solution. Treatment participants receive an embedded GPT-4.1 assistant; control participants complete the same task without it. All participants then answer incentivized follow-up questions about the task they had just completed without access to the assistant.

AI assistance substantially raises performance for both education groups, with larger gains among low-education participants. The overall task score increases by 1.242 standard deviations (SD) for low-education individuals and by 0.834 SD for high-education individuals. Without AI, high-education participants outperform low-education participants by 0.548 SD. With AI, the gap falls to 0.139 SD, closing 75% of the baseline difference. Performance remains dispersed among treated participants, indicating that the result is not mechanically driven by generalized convergence to a perfect score. Chat logs of participants' interactions with the assistant help interpret both the narrowing and persistence of the gap. Lower-education participants obtain substantial AI assistance, while higher-education participants give more detailed instructions, structure interactions more systematically, and integrate AI output more selectively.

Large gains in task performance do not by themselves distinguish productive assistance from delegation to the tool. Experimental evidence shows that gains from AI-assisted performance need not carry over to subsequent unassisted performance, which may even decline when AI substitutes for effortful problem-solving (Bastani et al., 2025; Shen and Tamkin, 2026; Liu et al., 2026; Autor et al., 2026). The non-AI-assisted follow-up allows us to examine whether treated participants perform worse once assistance is removed and whether any short-run carry-over is associated with participants' own engagement with the problem.

The follow-up results do not support a pure-delegation interpretation: once AI is removed, treated participants do not perform worse than controls. In fact, AI access modestly improves follow-up performance among low-education participants by 0.171 SD, while the estimated effect among high-education participants is smaller and statistically insignificant. Despite this modest carry-over, a sizable education gap re-emerges in the follow-up, consistent with underlying human capital continuing to shape unassisted performance. To further interpret these results, we divide treated participants in an exploratory analysis into four mutually exclusive groups based on AI assistance and task engagement. Intensive AI assistance is associated with strong task

[2]This classification was preregistered, and the results are robust to alternative education-group thresholds.

performance even when engagement is low, but follow-up performance is substantially higher only when intensive AI assistance is combined with high task engagement.

Our study builds on a rapidly growing literature on generative AI in labor-market settings. A common finding is that AI assistance raises average productivity and produces larger gains among workers with lower baseline performance or experience. In an online experiment with 444 college-educated professionals, Noy and Zhang (2023) find that ChatGPT increases output quality by about 0.45 SD and reduces completion time by about 0.83 SD, with larger gains for lower baseline performers. In a lab-in-the-field experiment with 758 consultants, Dell'Acqua et al. (2026) find substantial productivity and quality gains on tasks inside the "jagged technological frontier," with larger improvements for initially lower-performing consultants. Field deployments among customer-support agents and software developers likewise show sizable average gains that are concentrated among novice or less experienced workers (Brynjolfsson et al., 2025; Cui et al., 2026).[3] The pattern is not universal. Recent studies of software developers, entrepreneurs, and university debaters find settings in which gains are concentrated among more experienced or higher-ability users, or in which lower-baseline users do not benefit (Daniotti et al., 2026; Kreitmeir and Raschky, 2024; Otis et al., 2026; Roldan, 2024). Taken together, the evidence suggests that generative AI often compresses performance differences within relatively homogeneous groups, though heterogeneous effects can vary with the task, the users, and the capabilities of the technology. Benzell and Myers (2026) formalize how these elements can generate different, and potentially non-monotonic, effects on inequality as AI capabilities improve.

We make three related contributions. First, and most importantly, our experimental setup is explicitly designed to measure whether AI compresses performance gaps across groups with different levels of formal education. To the best of our knowledge, this is the first study to directly compare AI-assisted performance on the same task across education-based groups. Complementing evidence from firms and occupations, our design targets the between-education-group performance gap as the primary estimand, thereby shifting the focus from performance compression within relatively homogeneous worker populations to the effect of AI on productivity gaps across the education distribution.[4] Second, we distinguish contemporaneous AI-enabled performance from subsequent unassisted performance. The follow-up module allows us to assess whether the observed gains persist without AI or are entirely explained by temporary delegation. Third, we use the chat logs to characterize the types of AI use associated with the distributional pattern behind the main result. This lets us

[3]Studies within other firms and occupations report similar patterns (Chen and Chan, 2024; Choi et al., 2024; Kanazawa et al., 2026; Marsal and Perkowski, 2025); Imas (2026) reviews this evidence.

[4]Related work by Haslberger et al. (2025) studies a representative UK sample, but baseline performance does not differ by educational attainment. Our task generates baseline differentiation, allowing us to test whether AI narrows an education-based gap.

examine whether lower-education participants obtain substantial support from AI and how differences in use help account for why the education gap narrows but does not disappear.

2 Experimental design

2.1 Recruitment strategy

We conducted the online experiment in Argentina between September and November 2025. Participants were recruited through three panel survey companies and were required to complete the experiment on a personal computer. In addition to each panel's standard participation arrangements, participants could earn up to AR$25,000 (approximately US$18) in supermarket gift cards based on performance in the task and follow-up questions.[5] The experiment was designed to take about 20 minutes, with a maximum duration of one hour, and took 21 minutes on average.

We recruited low- and high-education level individuals aged 25–45 and aimed for broad balance by age and gender within each education group. As preregistered, low education comprises a completed high-school degree with no postsecondary education or completion of less than half of a university or non-university tertiary program; high education comprises completion of at least half of such a program. Standard education groupings often treat postsecondary education short of a bachelor's degree as an intermediate "some college" category (Goldin and Katz, 2008; Acemoglu and Autor, 2011). That grouping is especially heterogeneous in Argentina: non-university tertiary programs, including technical and professional tracks, typically last three years or more. Furthermore, public tertiary education is tuition-free and generally open admission, so many individuals enroll but complete only a limited portion of their studies. We therefore use completion of at least half of a postsecondary program as the threshold for the high-education group, yielding a more informative stratification of formal education. Section 7 shows that the results are robust to alternative education classifications.

2.2 Experiment structure

The experiment was administered through a browser-based platform. After consenting and completing a short demographic questionnaire, participants were randomly assigned by the platform to treatment or control. Both groups received the same

[5]Participants were informed that performance, evaluated on content and format, determined rewards ranging from AR$0 to AR$25,000. For context, the hourly minimum wage at this time was AR$ 1,610. For simplicity and fairness, we assigned participants to three equally sized payment tiers—AR$0, AR$12,500, or AR$25,000—according to relative performance within their treatment-by-education cell.

incentivized task. Treatment participants had access to an embedded GPT-4.1-based chat assistant; control participants did not.[6] Participants were instructed not to consult outside sources or leave the task page, and the platform recorded page exits. For treated participants, all exchanges with the assistant were recorded for subsequent analysis.

Before the task, treated participants completed a brief tutorial on interacting with the assistant, emphasizing the need to provide relevant instructions and context. The tutorial used an unrelated, deliberately simple practice exercise and did not teach participants how to analyze the substantially more complex experimental task. Among treated participants, 94% interacted with the assistant during the tutorial and 77% completed the tutorial module, with nearly identical completion across education groups.

The task presented a realistic hypothetical business problem through an email from the participant's boss and an attached report containing text, a figure, and a table.[7] Participants had to answer three diagnostic questions and propose a solution in an email response. The task was self-contained and did not require industry- or firm-specific knowledge. It combined reading and data comprehension, reasoning, creative problem-solving, and writing. We designed three parallel versions, set in a cafeteria, a food-delivery service, and a home-appliances store, with participants randomly assigned to one version. The versions were piloted to be challenging for low-education participants while remaining understandable and completable without AI, so that we could observe meaningful baseline human-capital differences without strong ceiling or floor effects (i.e., minimum scores for most low-education participants or perfect scores for high-education ones). Consistent with that design, low-education controls reported greater difficulty, very few described the task as very difficult, and most reported at least some familiarity with comparable activities (Appendix Figures A.1-A.2).

Immediately after submitting the task, participants completed an incentivized follow-up module without access to the assistant. The first item asked them to articulate the root cause as if responding to a boss who had not read their email. Two multiple-choice items then assessed interpretation of the task figure and recall of a factual detail. The module measures short-run unassisted performance on related content after AI exposure, not durable skill formation on a new task. Appendix B.1 depicts the different stages of the experiment. Appendix F reproduces the recruitment emails, consent language, instructions, tutorial, all task versions, follow-up questions, and the prompts supplied to the assistant.

[6] At the time of fieldwork, Qualtrics did not support the required AI integration, so we developed a custom module that embedded the assistant, recorded participant-assistant interactions, and tracked page exits.

[7] For treated participants, the assistant was preloaded with the attached report but not the boss's email or task instructions, and initialized with a basic system prompt, reproduced in Appendix F.7.

2.3 Outcomes of interest

The main outcome is task performance, graded on a preregistered 0–10 scale. The overall score weights content two-thirds and writing one-third. For content, participants receive up to six points for diagnosis (two for each of the three questions in the boss's email) and up to four points for the proposed solution. The solution receives credit for addressing the root cause, offering a concrete proposal, and being realistic; if it does not address the root cause, the entire solution component receives zero. The writing score allocates points to spelling and grammar, clarity, organization, and tone. To avoid giving high writing scores to very short incorrect responses, a preregistered deterministic rule assigns zero writing points to responses under 200 characters with zero content points. In addition to the overall score, we report content and writing separately, as well as an indicator for correctly identifying the root cause of the problem.

Other preregistered outcomes are task speed, perceived difficulty, and follow-up performance. Task-completion time is measured in minutes and top-coded at 20. Perceived difficulty is recorded on a five-point scale. The overall follow-up score assigns one-half weight to the open-ended item and one-quarter to each multiple-choice item, each scored as correct or incorrect. The open-ended question is scored on a 0–2 scale. We also report the open-ended and multiple-choice components separately. All scores are standardized using the low-education control group's mean and standard deviation. The overall task score is the natural focal outcome; the content and writing measures are direct components, while duration and follow-up performance are related but distinct outcomes, following the logic of hierarchical outcome families in Calónico and Galiani (2025).

We developed task-specific rubrics from the preregistered scoring criteria and used OpenAI's GPT-5-mini model to grade the task and open-ended follow-up responses according to these rubrics. Grading is performed one response at a time.[8] The model receives the relevant rubric and the participant's response and returns item-level subscores, which are then mechanically aggregated. Because LLM scores vary slightly across runs, we grade each response ten times and use the average score.[9] Section 7 shows that the results are robust to alternative grading procedures.

[8] Rubric-based LLM grading has shown high reliability in related scoring and text-annotation tasks (Pack et al., 2024; Gilardi et al., 2023; Bermejo et al., 2025).

[9] Appendix D documents the grading procedure and provides a link to the full prompts and task-specific rubrics in the public repository.

3 Sample, take-up, and estimation strategy

3.1 Sample and balance in baseline characteristics

Among the 1,795 participants who met the demographic criteria, treatment and control groups were jointly balanced in baseline characteristics. The experiment was completed by 1,174 participants (65%), who constitute the analysis sample. Stantcheva (2023) reports non-completion rates from 15% to somewhat above 30% in online survey experiments; our 35% rate is slightly above that range, consistent with the demands of a job-like problem-solving and writing task.

Attrition was not differential by treatment assignment. Completion rates were 64% in treatment and 66% in control ($p = 0.373$), with no significant differences within education groups: the treatment-control difference was 0.3 percentage points (pp) for low-education participants and 3.7 percentage points for high-education participants. Equality across all four treatment-by-education cells cannot be rejected. Moreover, baseline balance from the original randomization is preserved in the analysis sample, completers and non-completers are similar on observables within each education group, and treatment and control remain jointly balanced within education groups among completers.

The final sample contains 520 low-education and 654 high-education participants,[10] with an average age of about 35 and near gender balance in both groups. Employment is 78% among low-education participants and 89% among high-education participants, who also report about one additional year of work experience. Among low-education participants, 64% have only a high-school diploma and 36% completed less than half of a postsecondary degree. Among high-education participants, 29% completed at least half of a postsecondary degree, 63% completed one, and 8% have a graduate degree. Prior AI use is 68% and 77%, respectively. To assess comparability with the population of interest, we use Argentina's Household Survey (*Encuesta Permanente de Hogares*) and construct an education definition available in both datasets: secondary school as the highest completed level versus at least some postsecondary education. Within those groups, the experimental sample is broadly similar to the corresponding population in age, gender, educational attainment, and labor-market status. Appendix Tables A.1–A.5 and Figures A.3-A.4 report balance, sample composition, attrition comparisons, and the population comparison.

[10] Appendix B.2 documents the construction of this sample from the initial pool of respondents. Our sample is moderately unbalanced by education (44% low- and 56% high-education), partly because the personal-computer requirement was more limiting for low-education participants. This does not affect our main estimand, which is the difference in treatment effects across education groups.

3.2 Treatment take-up and compliance

Use of the embedded assistant was optional. Treated participants were encouraged to use it but told that rewards depended only on response quality, regardless of whether they used the tool. Take-up was high: 84% used the assistant during the task, including 81% of low-education and 86% of high-education participants. This difference is small relative to the gap in estimated treatment effects, so the main result is not generated by markedly different treatment-group take-up.

A potential concern is external AI use in the control group. Thirty-one percent of controls exited the task page, compared with 19% of treated participants. All participants were instructed not to consult external sources or seek help, and page exits triggered a reminder. The platform also saved snapshots of participants' responses every minute. We detect AI use among controls who left the page using two sources of evidence. First, we searched snapshots and final responses for direct indicators such as excerpts of an AI conversation, emojis, or characteristic bullet formatting. Second, we measured the largest increase in word count between consecutive snapshots and flagged unusually large jumps comparable with those of treated assistant users. Direct evidence identifies 6% of controls; the snapshot procedure identifies an additional 7%. Appendix B.3 describes the detection procedure in detail.

The combined procedure classifies 13% of controls as possible AI users, 11% among low-education and 15% among high-education participants.[11] The estimated first-stage differences in use between treatment and control are consequently almost identical, at 70 and 71 pp (Appendix Figure A.5). Our main estimates are intent-to-treat effects of access rather than effects of use. External AI use in the control group attenuates these effects relative to a counterfactual of zero AI access. The incomplete take-up also foreshadows an external-validity point discussed in the conclusion: our experiment reduces adoption frictions substantially but does not eliminate them.

3.3 Estimation strategy

The design targets the change in the performance gap between low- and high-education participants. For each preregistered outcome, we estimate

$$Y_i = \beta_0 + \beta_1 \textit{High education}_i + \beta_2 \textit{Treatment}_i + \beta_3 \textit{High education}_i \times \textit{Treatment}_i + u_i. \quad (1)$$

$\textit{High education}_i$ is an indicator variable equal to 1 if the participant is in the high-education group (0 if the participant has low education), and $\textit{Treatment}_i$ is an indicator variable equal to 1 if the participant was assigned to the treatment group and therefore

[11] Using Pangram, an AI-detection system, yields very similar estimates: 11% of low-education and 16% of high-education controls are classified as AI users.

had access to the virtual assistant while performing the task. β_0 is the low-education control mean and β_1 is the education gap without AI. The treatment effects for low- and high-education participants are β_2 and $\beta_2 + \beta_3$, respectively. Our coefficient of interest, β_3, is the effect of access to AI on the high- versus low-education gap. A negative value indicates convergence. We report the four implied cell means and economically relevant contrasts rather than only the regression coefficients. Heteroskedasticity-robust standard errors are used because assignment occurred at the individual level.[12]

4 Impact of AI assistance on gaps in task performance

Figure 1 presents the central result, while Table 1 reports results for all task and follow-up outcomes. The figure's left panel reports mean standardized scores in the four treatment-by-education cells, while the right panel reports the intent-to-treat effect within each education group. Because scores are standardized relative to the low-education control group, its mean is zero by construction.

Without AI, the task generates the intended differentiation by formal education: high-education participants score 0.548 SD above low-education participants. Access to the assistant produces large gains for both groups. The overall score rises by 1.242 SD for low-education participants and 0.834 SD for high-education participants, with both estimates significant at the 1% level. The difference in treatment effects is −0.408 SD and also significant at the 1% level. The interaction remains significant after a Romano–Wolf adjustment (Romano and Wolf, 2005) across the conceptually distinct preregistered outcomes of overall task performance, duration, and overall follow-up performance ($p = 0.002$). Thus, access to the assistant closes 75% of the baseline education gap (reducing it from 0.548 to 0.139 SD) while raising performance in both groups.

The effects are large enough that low-education participants with access to AI outperform high-education controls on average. Access therefore more than offsets the baseline disadvantage in this task. Convergence remains incomplete: among treated participants, high-education individuals still score 0.139 SD higher, although the estimate is marginally insignificant ($p = 0.110$). This difference is consistent with AI substituting for some inputs scarcer among low-education participants without eliminating the role of human capital in deciding what to ask, evaluating the generated response, and integrating it into the final answer.

Figure 2 shows that the mean result is not driven by a pile-up at the upper bound. The assistant shifts both distributions to the right, but scores remain dispersed: only 13% of treated low-education participants and 21% of treated high-education

[12] An alternative estimand is the effect of actual AI use, instrumented by randomized access. Our primary question, however, is about the effect of access; because the first stage is very similar across education groups, our conclusions about differential effects are unchanged.

participants reach the top category (scores from 9.75 to 10 on the unstandardized 0–10 scale). The control distributions likewise show substantial variation and no strong floor effect. The differential treatment effect therefore reflects broad improvement rather than mechanical convergence to a perfect score.

Additional task outcomes. For content and writing scores, results are very similar in terms of baseline gaps, treatment effects, differential effects by education, and remaining gaps among treated participants. Root-cause identification follows the same pattern, rising from 49.5% to 84.1% among low-education participants and from 64.7% to 88.1% among high-education participants, with the treatment effect 11.2 pp larger for the low-education group and significant at the 5% level. Thus, the equalizing effect is not driven solely by writing quality, arguably the easiest margin for AI to improve, but also appears in content and root-cause diagnosis.[13] AI reduces completion time in both groups, with no significant differential effect by education, making duration the only task outcome without an equalizing pattern. Appendix C.1 reports the full results.

5 Performance after AI is removed

5.1 Carry-over of gains in the follow-up module

The non-AI-assisted follow-up helps distinguish gains that are purely contemporaneous with access from gains that partly carry over once the tool is unavailable. The outcome should nevertheless be interpreted narrowly: it measures short-run unassisted understanding and recall of the task just completed, not durable skill acquisition or performance on a new task.

Baseline follow-up performance again differs by education: among controls, high-education participants score 0.300 SD above low-education participants. Prior AI access increases the follow-up score of low-education participants by 0.171 SD, significant at the 5% level. The high-education estimate is 0.071 SD and statistically insignificant.[14] Their difference, −0.100 SD, is imprecisely estimated and indistinguishable from zero.

The results have two implications. First, neither education group performs worse after AI is removed. This is inconsistent with the strongest pure-delegation account, in which access improves task performance at the expense of subsequent unassisted performance. The positive effect on low-education participants also indicates that at

[13] Perceived difficulty also declines with AI access in both education groups, with no differential effect by education; education-based differences nevertheless remain among treated participants (Appendix Figure A.1).

[14] The treatment effect for low-education participants is concentrated in the open-ended question asking them to articulate the root cause, while neither education group shows an effect on the multiple-choice items (see Appendix C.2 for further details).

least part of the contemporaneous gain carries over. Second, carry-over is modest relative to the baseline education difference: the treated-group follow-up gap remains 0.200 SD and significant at the 5% level. Although the imprecisely estimated interaction prevents us from concluding that AI compresses unassisted differences, the persistence of a sizable follow-up gap is consistent with underlying human capital continuing to matter once assistance is unavailable.

5.2 Engagement, delegation, and carry-over of gains

To further interpret the follow-up, we conduct an exploratory analysis among treated participants using two behavioral dimensions during the main task. The first captures exposure to AI-generated solutions, using the share of task components for which a participant asked the assistant for help. The second captures participants' own engagement with the task, proxied by time spent on the task. Splitting each measure at the treatment-group median yields four mutually exclusive groups: high engagement/high assistance, low engagement/high assistance, high engagement/low assistance, and low engagement/low assistance. The groups have similar shares across education categories.

Table 2 relates the four categories to task and follow-up scores, using low engagement/low assistance as the omitted category. Because engagement and assistance are post-treatment choices, these comparisons are descriptive associations, not causal effects of assistance or engagement. High AI assistance is associated with strong task performance at both engagement levels. Pooling low- and high-education participants, task scores are 1.878 SD for high-engagement/high-assistance participants and 1.972 SD for low-engagement/high-assistance participants, a statistically insignificant difference. The same qualitative pattern appears within both education groups.

The pattern differs in the follow-up. Scores are 0.732 SD for the high-engagement/high-assistance group but only 0.065 SD for the low-engagement/high-assistance group ($p < 0.01$ for the difference), despite their similar main-task scores. Follow-up performance is also high among participants with high engagement but low assistance.[15] Overall, these patterns suggest that AI assistance alone can generate strong task performance, but carry-over is concentrated among participants with high task engagement.

[15]The differences are quantitatively similar after controlling for education, age, gender, employment, and work experience. Using alternative cutoffs (33rd and 66th percentiles) to define high versus low assistance and engagement yields similar results.

6 Patterns of AI use

We next use an LLM-based procedure to classify the recorded conversations of the 471 treated participants who interacted with the assistant during the task. We characterize their behavior along three dimensions. The first concerns how participants work through the task with AI, including the share of task components for which they request help and the detail of instructions used to guide the assistant's reasoning. The second concerns how they generate and use the final output, including drafting requests, stylistic or substantive constraints, and whether AI-generated text is copied fully, copied partly, paraphrased, or not used. The third concerns how they orchestrate the interaction, including the specificity of the initial prompt, whether they iterate, and whether they organize the conversation across substantive steps. Since these measures are post-treatment choices, they provide descriptive evidence to characterize AI use and contextualize the residual performance gap, but they do not represent estimates of causal effects of particular prompting strategies. Appendix E provides complete definitions and points to the classification prompt in the public repository.

Appendix Table A.7 shows that the AI-use measures are positively associated with task scores, and jointly explain about 49% of their variation. These associations are descriptive rather than causal: a behavior can be associated with better output because it is productive, because more capable participants choose it, or both.

Table 3 summarizes AI-use patterns by education group. Interaction intensity is very similar across groups: both send nearly three messages; low-education users send 794 characters on average and high-education users 847. They request assistance for roughly 61% of task components, and frequently ask the assistant to generate the response directly. About two-thirds of users make an explicit drafting request. These common patterns are central to understanding the large convergence: lower-education participants obtain substantial assistance with diagnosis, solution design, and written production.

Differences emerge more clearly in qualitative dimensions. High-education participants provide more detailed instructions aimed at guiding the assistant's reasoning, scoring 0.174 SD higher on the corresponding index. Higher-education users also provide somewhat more detailed drafting instructions, scoring 0.066 SD higher on the corresponding index, are 6.7 pp more likely to begin with a highly specific prompt, and are 7.5 pp more likely to use a structured workflow. Iterative engagement, by contrast, is almost identical across groups. These patterns suggest differences in how the interaction is organized rather than on how often the assistant is used. They may partly reflect experience: prior AI use is 68% among low-education participants and 77% among high-education participants. At the same time, reported usefulness of the assistant and comfort with its use are very similar across education groups (Appendix

Figure A.6), suggesting that these differences are not driven by lower comfort with the tool among low-education participants.

The groups also differ in the use of AI-generated text. Most participants rely heavily on AI content, but low-education users are 10.1 pp more likely to copy the generated output in full. High-education users are more likely to incorporate the output partially or paraphrase it. This does not imply that full copying is itself harmful or that selective integration causally raises performance. It does indicate that higher-education participants more often treat the output as an input into their own production process, whereas lower-education participants more often use the assistant as a near-complete substitute for drafting.

Taken together, the logs suggest that lower-education participants obtain substantial assistance, helping explain the large narrowing of the gap, while higher-education participants use AI more effectively, helping explain why the gap does not disappear.

7 Outcome validation and robustness

The central result is stable across several validation and robustness checks. First, using any individual GPT-5-mini grading iteration instead of the average produces virtually the same treatment effects and differential effect; pairwise correlations across iterations range from 0.981 to 0.986.

Second, two economics students manually graded a 10% random sample of responses using the same criteria, blind to treatment assignment, education group, and the hypothesis. They were told not to view unusually strong answers as suspicious or downgrade them on that basis. The correlation between the main AI-based score and the average human score is above 0.9.

Third, regrading every response with Anthropic's Claude Haiku 4.5 and Google's Gemini Flash 3.0 produces a differential treatment effect across education groups very close to our main estimate.

Fourth, an independent researcher also developed three procedures from the preregistration and task materials, without our detailed correction guidelines or treatment and education indicators. Two use independently generated rubrics; the third ranks responses through pairwise comparisons and aggregates them using Elo ratings. The first two scores correlate above 0.9 with our main score. The Elo score has a Pearson correlation of 0.784 and a rank correlation of 0.934. All three reproduce the differential treatment pattern.[16]

[16]Twenty-nine responses were inadvertently not sent for independent grading. The conclusions are unchanged when the missing scores are imputed using the minimum, maximum, mean, or median within the corresponding treatment-by-education cell.

Lastly, the estimates are also stable after adding task fixed effects and baseline controls, excluding 25 very low-quality responses, excluding participants with incomplete postsecondary education, or using alternative education-group definitions. Appendix D documents the AI-assisted grading procedures and validation, and Appendix Figures A.7–A.9 summarize the robustness checks.

8 Conclusion

This paper asks whether generative AI narrows education-based productivity gaps and whether its gains persist once the tool is removed. Our experimental results indicate that AI raises performance in both high and low education groups and closes three-quarters of the baseline gap, with larger gains for low-education participants. The follow-up results help interpret these gains: they do not support a pure-delegation account, as treated participants do not perform worse once AI is removed and part of the gain carries over for low-education participants. Moreover, follow-up performance is substantially higher when intensive AI assistance is combined with sustained task engagement. Finally, the chat logs help interpret why convergence is incomplete, as low-education participants obtain substantial assistance from AI, while high-education participants request similar levels of assistance but use the tool somewhat more effectively.

Our interpretation emphasizes the distinction between underlying human capital and effective productivity in task execution. Under this view, formal education reflects accumulated inputs such as abstract reasoning and experience with complex written communication. AI can substitute for some of these inputs when they are scarcer among lower-education participants by helping to structure the problem, formulate the response, and produce the final output. It can therefore reduce the effective expertise required to complete a complex but general task, consistent with recent task-based work (Autor and Thompson, 2025; Althoff and Reichardt, 2026; Danieli, 2026; Hosseini and Lichtinger, 2026) and with the potential of AI as an "expertise-leveling" technology (Acemoglu et al., 2026; Agrawal et al., 2024; Freund and Mann, 2025). At the same time, the experiment does not show that AI eliminates underlying human-capital differences. The task still requires participants to evaluate, select, and integrate AI-generated material, and the remaining education gap in AI-assisted performance together with the differences in use patterns are consistent with human capital continuing to matter. This combination of substantial convergence without complete equalization is consistent with the argument in Autor (2024) that AI can broaden access to expertise and with the expertise-leveling framework of Acemoglu et al. (2026).

The influential "jagged technological frontier" framework developed by Dell'Acqua et al. (2026) provides a useful organizing principle for the scope of this result. Our task

lies in a part of that frontier where current systems can provide substantial assistance on complex knowledge work. The same framework cautions against extrapolating the equalizing pattern to tasks outside the frontier, including tasks that rely more heavily on interpersonal coordination or capabilities that current AI does not replicate well.

The findings are also specific to a particular generation of AI capabilities. As capabilities expand, the relative gains of low- and high-education users may shift, potentially attenuating or reversing the equalizing pattern; Benzell and Myers (2026) formalize why distributional effects need not evolve monotonically.

Finally, the experiment embeds the assistant, provides a tutorial, and assigns access. Our estimates are therefore conditional on availability and unusually low adoption frictions. Whether the capability gains documented here translate into narrower labor-market inequality depends on who receives access, who adopts the tool, and the organizational conditions under which it is used (Bursztyn et al., 2026; Almog, 2025; Imas and Shukla, 2026; Mehmood et al., 2026), and more broadly on whether experimental effects scale to policy-relevant settings (Al-Ubaydli et al., 2020). We consequently do not interpret the estimates as direct predictions about wages or aggregate equilibrium inequality. Rather, our results show that under broad access and low adoption frictions, generative AI has the potential to substantially narrow education-based differences in workplace-style task performance.

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Tables and Figures

Table 1: Main results

	Task					
	Overall (z-score)	Writing (z-score)	Content (z-score)	Detected root cause (0/1)	Duration (minutes)	Follow-up (z-score)
Low education						
$\overline{Y}_L^C$	0.000	0.000	0.000	0.495***	10.697***	0.000
	(0.060)	(0.060)	(0.060)	(0.029)	(0.367)	(0.060)
$\overline{Y}_L^T$	1.242***	1.201***	1.166***	0.841***	9.736***	0.171***
	(0.065)	(0.059)	(0.066)	(0.022)	(0.397)	(0.063)
$\overline{Y}_L^T - \overline{Y}_L^C$	1.242***	1.201***	1.166***	0.346***	-0.961*	0.171**
	(0.088)	(0.084)	(0.089)	(0.036)	(0.541)	(0.087)
High education						
$\overline{Y}_H^C$	0.548***	0.503***	0.525***	0.647***	10.423***	0.300***
	(0.061)	(0.056)	(0.062)	(0.025)	(0.319)	(0.058)
$\overline{Y}_H^T$	1.382***	1.323***	1.302***	0.881***	8.909***	0.371***
	(0.059)	(0.054)	(0.059)	(0.018)	(0.339)	(0.056)
$\overline{Y}_H^T - \overline{Y}_H^C$	0.834***	0.820***	0.777***	0.234***	-1.514***	0.071
	(0.084)	(0.078)	(0.085)	(0.031)	(0.465)	(0.080)
High vs. Low education						
$\overline{Y}_H^C - \overline{Y}_L^C$	0.548***	0.503***	0.525***	0.152***	-0.274	0.300***
	(0.086)	(0.082)	(0.086)	(0.038)	(0.486)	(0.083)
$\overline{Y}_H^T - \overline{Y}_L^T$	0.139	0.122	0.136	0.040	-0.827	0.200**
	(0.087)	(0.080)	(0.088)	(0.028)	(0.522)	(0.084)
$(\overline{Y}_H^T - \overline{Y}_H^C) - (\overline{Y}_L^T - \overline{Y}_L^C)$	-0.408***	-0.382***	-0.389***	-0.112**	-0.553	-0.100
	(0.122)	(0.114)	(0.123)	(0.048)	(0.713)	(0.118)
Observations	1174	1174	1174	1174	1174	1174

Note: The first three columns report the results of estimations where the dependent variable is the task score (average across 10 grading iterations), standardized relative to the low-education control group: overall score (column 1), writing score (column 2), and content score (column 3). In column 4, the dependent variable is the average across ten grading iterations of a dummy for whether the respondent detected the root cause of the problem. In column 5, the dependent variable is time taken to complete the task (in minutes). In the last column, the dependent variable is the overall score in the follow-up questions, standardized relative to the low-education control group. The top and central panels show the means of the control and treatment groups and the difference between the treatment and control group for the low- and high-education groups, respectively. The bottom panel shows the difference between control and treatment means for the high- and low-education groups, as well as the difference between the treatment effects of both education groups. Because the overall, content, and writing outcomes are standardized separately, the standardized overall gap need not lie between the two component gaps. * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 2: Relationship between scores, AI assistance and task engagement

	Task (z-score)			Follow-up (z-score)		
	Pooled	Low education	High education	Pooled	Low education	High education
High engagement + High assistance	1.324***	1.286***	1.351***	0.886***	0.743***	0.992***
	(0.098)	(0.145)	(0.132)	(0.106)	(0.168)	(0.134)
Low engagement + High assistance	1.418***	1.342***	1.469***	0.219	0.254	0.193
	(0.097)	(0.148)	(0.129)	(0.112)	(0.176)	(0.146)
High engagement + Low assistance	0.372**	0.244	0.502**	0.644***	0.591***	0.724***
	(0.118)	(0.169)	(0.165)	(0.114)	(0.170)	(0.154)
Constant	0.554***	0.555***	0.553***	-0.154*	-0.235	-0.096
	(0.082)	(0.122)	(0.110)	(0.077)	(0.122)	(0.100)
Observations	562	244	318	562	244	318
R^2	0.356	0.356	0.359	0.130	0.091	0.172
P-values						
High/High=Low/High	0.206	0.629	0.229	0.000	0.005	0.000
High/High=High/Low	0.000	0.000	0.000	0.029	0.357	0.069
Low/High=High/Low	0.000	0.000	0.000	0.000	0.055	0.001

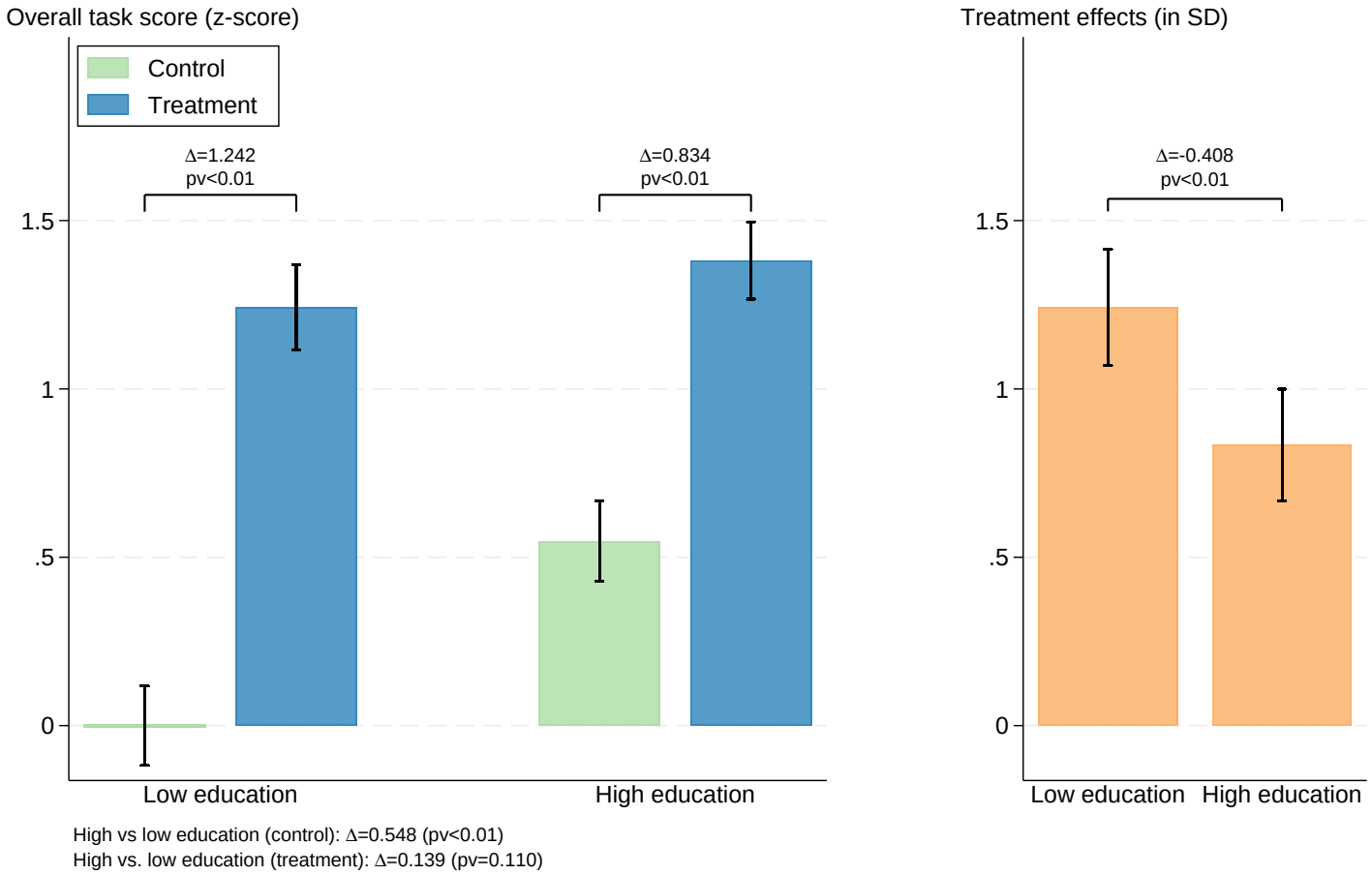
Note: This table reports the results of estimations where the dependent variable is the overall score in the task (columns 1-3) and follow-up questions (columns 4-6), standardized relative to the low-education control group. We conduct these regressions pooling low- and high-education participants, as well as separately for each group, as specified in the column headers. The regressors are indicator variables for four mutually exclusive groups defined by whether participants spent above- or below-median time on the task (engagement) and whether their requested AI solution assistance was above or below the median. Task time is measured relative to the treatment-group median. AI solution assistance is measured as the share of task components for which the participant requested help from the AI assistant. The omitted category is participants with below-median time on the task and below-median AI solution assistance. * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 3: AI use by education level

	Low education	High education	Difference
Number and length of messages sent			
Number of messages sent to AI assistant	2.813	2.971	0.158
Total characters sent to AI assistant	794.172	847.396	53.224
Assistance with working through the task			
Share of task components with any AI assistance	0.608	0.615	0.007
Detailed instructions for thinking through task (std. index)	-0.000	0.174	0.174*
Assistance with generating the output			
Explicitly requested AI to draft answer	0.702	0.652	-0.050
Requested AI to extend or complete user output	0.020	0.040	0.020
Requested AI to edit user output	0.035	0.022	-0.013
No explicit request for AI assistance for output generation	0.242	0.286	0.043
Detailed instructions for drafting response (std. index)	-0.000	0.066	0.066
Use of AI output			
Final output full copy/paste of AI content	0.702	0.601	-0.101**
Final output partial copy/paste of AI content	0.242	0.308	0.065
Final output paraphrases AI content	0.045	0.073	0.028
Final output does not rely on AI content	0.010	0.018	0.008
Workflow/orchestration of the conversation			
High initial specificity	0.288	0.355	0.067
Iterative engagement	0.465	0.473	0.008
Structured workflow	0.141	0.216	0.075**
Observations	198	273	

Notes: This table reports mean values of AI-use measures for low- and high-education participants who used the assistant to complete the task. Column 3 reports the difference between groups. Measures are constructed from detailed interaction logs. Assistance with working through the task captures the share of task components for which participants requested AI help and an index of disciplined, evidence-based reasoning instructions and relevant contextual information. Output-generation measures classify drafting requests and an index of style- and content-specific constraints. Use of AI output classifies full copy-paste, partial copy-paste, paraphrasing, or no direct reliance. Workflow measures capture initial specificity, iterative engagement, and structured workflow. Indices are standardized relative to the low-education group. See Appendix E. * significant at 10%; ** significant at 5%; *** significant at 1%.

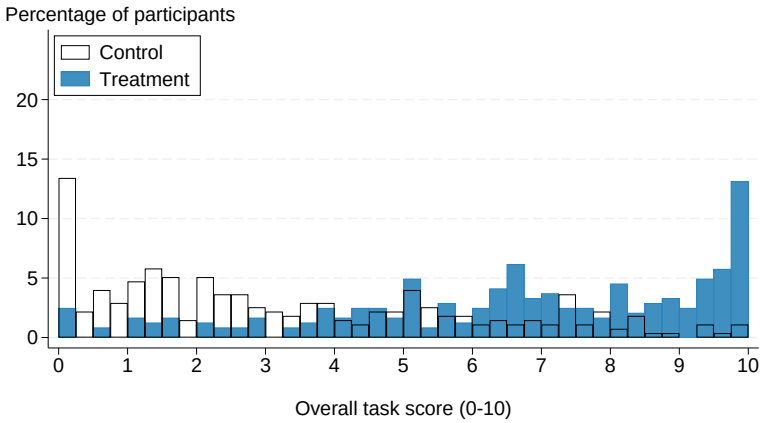
Figure 1: Effects of AI assistance on overall task score by education level



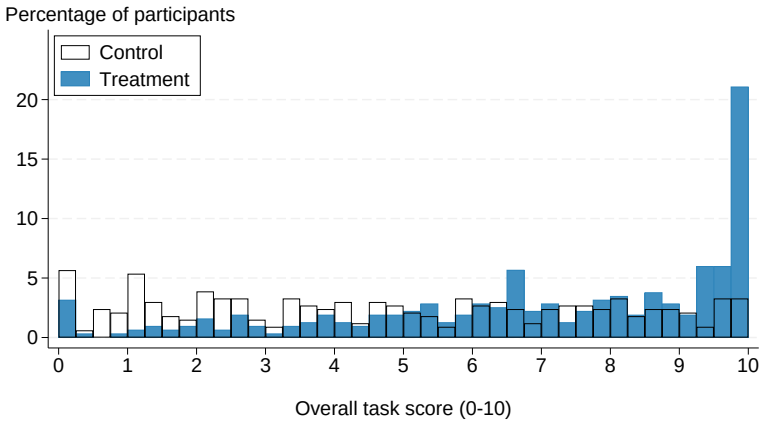
Notes: The left panel shows the average score on the task (average across 10 grading iterations), standardized relative to the low-education control group, along with 95% confidence intervals, separately for high- and low-education participants in the treatment and control groups. Above each education group, we report the difference between the treatment and control groups within that education category. The right panel presents the estimated treatment effects for high- and low-education participants. Above, we report the difference between the treatment effects of high- and low-education participants.

Figure 2: Distribution of overall task score by treatment and education group

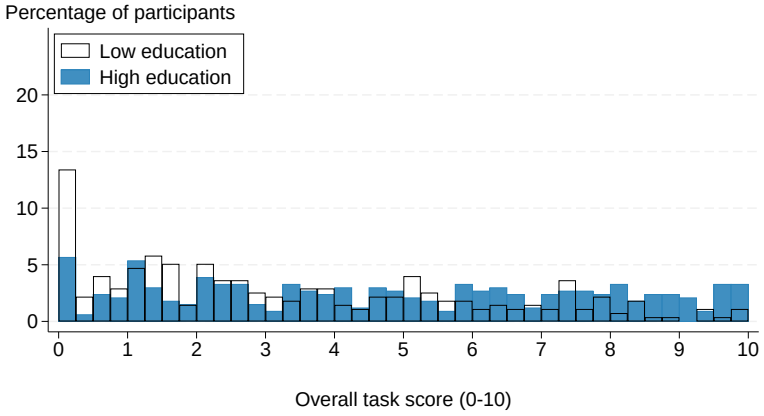
(a) Low-education sample: treatment vs. control



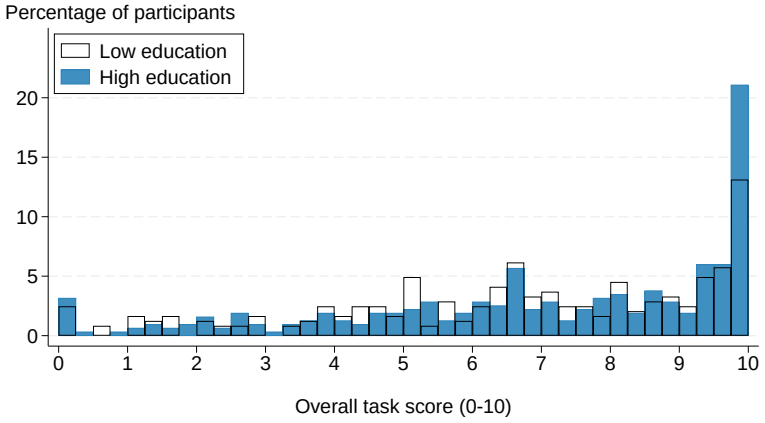
(b) High-education sample: treatment vs. control



(c) Control group: high- vs. low-education



(d) Treatment group: high- vs. low-education



Notes: The top figures show the distribution of overall task score (on a 0 to 10 scale, prior to standardizing) by treatment group for the sample of low-education (left) and high-education individuals (right). The bottom figures show the analogous distribution by education group for the sample of control group (left) and treatment group individuals (right).

Online Appendix for: "Does generative AI narrow education-based productivity gaps? Evidence from a randomized experiment"

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A Appendix tables and figures

Table A.1: Balance between treatment and control

Variable	Control	Treatment	Difference	P-value (diff.)
Panel A: Initial sample				
Age	35.234	35.239	0.005	0.985
Female	0.511	0.501	-0.010	0.664
Male	0.485	0.493	0.008	0.743
Another gender	0.004	0.007	0.003	0.474
High school	0.311	0.249	-0.063***	0.003
Less than half postsecondary	0.140	0.171	0.031*	0.072
More than half postsecondary	0.151	0.170	0.019	0.279
Complete postsecondary	0.349	0.367	0.017	0.445
Graduate degree	0.049	0.045	-0.004	0.678
Self employed or employer	0.155	0.136	-0.019	0.259
Employee	0.692	0.702	0.010	0.638
Not employed	0.153	0.162	0.009	0.618
Years of work experience	10.980	11.257	0.276	0.376
Observations	922	873		
P-value (joint significance)				0.172
Panel B: Sample that finished experiment				
Age	35.124	34.945	-0.179	0.601
Female	0.490	0.504	0.013	0.648
Male	0.503	0.491	-0.012	0.677
Another gender	0.007	0.005	-0.001	0.789
High school	0.306	0.260	-0.046*	0.082
Less than half postsecondary	0.145	0.174	0.029	0.177
More than half postsecondary	0.154	0.167	0.014	0.525
Complete postsecondary	0.346	0.359	0.013	0.641
Graduate degree	0.049	0.039	-0.010	0.410
Self employed or employer	0.144	0.141	-0.003	0.875
Employee	0.696	0.705	0.009	0.750
Not employed	0.160	0.155	-0.005	0.803
Years of work experience	10.853	11.215	0.362	0.341
Observations	612	562		
P-value (joint significance)				0.616

Notes: This table shows the average characteristics at baseline of the control (column 1) and treatment (column 2) groups. Column 3 reports the difference between both groups, and column 4 reports the p-value of the difference in means test. In Panel A, the analysis is conducted for the entire sample, and in Panel B, for the sample that finished the experiment.

Table A.2: Comparison of participants who finished and did not finish the experiment

	Finished experiment			
	No	Yes	Difference	P-value (diff.)
Panel A: Low education				
Age	35.767	34.852	-0.915**	0.039
Female	0.557	0.494	-0.063*	0.096
Male	0.439	0.502	0.063*	0.096
Another gender	0.004	0.004	0.000	0.995
High school	0.653	0.640	-0.012	0.734
Less than half postsecondary	0.347	0.360	0.012	0.734
Self employed or employer	0.137	0.150	0.013	0.634
Employee	0.611	0.633	0.022	0.551
Not employed	0.252	0.217	-0.035	0.286
Years of work experience	10.626	10.310	-0.316	0.548
Observations	262	520		
P-value (joint significance)				0.260
Panel B: High education				
Age	35.499	35.187	-0.312	0.418
Female	0.499	0.498	-0.000	0.997
Male	0.496	0.494	-0.002	0.953
Another gender	0.006	0.008	0.002	0.690
More than half postsecondary	0.276	0.287	0.012	0.692
Complete postsecondary	0.635	0.633	-0.002	0.948
Graduate degree	0.089	0.080	-0.010	0.601
Self employed or employer	0.164	0.136	-0.028	0.234
Employee	0.749	0.754	0.005	0.874
Not employed	0.086	0.110	0.024	0.218
Years of work experience	11.760	11.596	-0.164	0.697
Observations	359	654		
P-value (joint significance)				0.913

Notes: This table compares the average characteristics at baseline of participants who did not complete and who completed the entire experiment. Column 3 reports the difference between both groups, and column 4 reports the p-value of the difference in means test. Panel A presents statistics for the low-education group, and Panel B for the high-education group.

Table A.3: Balance between treatment and control by education level – sample that finished the experiment

	Control	Treatment	Difference	P-value (diff.)
Panel A: Low education				
Age	34.797	34.914	0.117	0.821
Female	0.478	0.512	0.034	0.440
Male	0.518	0.484	-0.035	0.433
Another gender	0.004	0.004	0.000	0.931
High school	0.678	0.598	-0.079*	0.061
Less than half postsecondary	0.322	0.402	0.079*	0.061
Self employed or employer	0.149	0.152	0.003	0.922
Employee	0.627	0.639	0.013	0.768
Not employed	0.225	0.209	-0.016	0.667
Years of work experience	9.931	10.738	0.807	0.176
Observations	276	244		
P-value (joint significance)				0.436
Panel B: High education				
Age	35.393	34.969	-0.424	0.355
Female	0.500	0.497	-0.003	0.936
Male	0.491	0.497	0.006	0.883
Another gender	0.009	0.006	-0.003	0.698
More than half postsecondary	0.280	0.296	0.016	0.655
Complete postsecondary	0.631	0.635	0.004	0.910
Graduate degree	0.089	0.069	-0.020	0.341
Self employed or employer	0.140	0.132	-0.008	0.771
Employee	0.753	0.755	0.002	0.959
Not employed	0.107	0.113	0.006	0.805
Years of work experience	11.610	11.582	-0.028	0.954
Observations	336	318		
P-value (joint significance)				0.959

Notes: This table shows the average characteristics at baseline of the control (column 1) and treatment (column 2) groups, for the sample that finished the experiment. Column 3 reports the difference between both groups, and column 4 reports the p-value of the difference in means test. Panel A presents statistics for the low-education group, and Panel B for the high-education group.

Table A.4: Descriptive statistics by education group

	Low education	High education
Demographics		
Age	34.852	35.187
Female	0.494	0.498
Male	0.502	0.494
Another gender	0.004	0.008
Educational attainment		
High school	0.640	0.000
Less than half postsecondary	0.360	0.000
More than half postsecondary	0.000	0.287
Complete postsecondary	0.000	0.633
Graduate degree	0.000	0.080
Labor market characteristics		
Self employed or employer	0.150	0.136
Employee	0.633	0.754
Not employed	0.217	0.110
Years of work experience	10.310	11.596
Observations	520	654

Notes: This table shows the average characteristics of the low- and high-education participants.

Table A.5: Comparison of the experimental sample with the Household Survey

	Experimental sample	Household Survey
Panel A: Low education (complete high school)		
Age	35.778	34.657
Female	0.456	0.501
Male	0.544	0.499
Another gender	0.000	-
Self employed	0.153	0.189
Employee	0.646	0.591
Not employed	0.201	0.216
Panel B: High education (at least some postsecondary education)		
Age	34.746	35.255
Female	0.512	0.574
Male	0.479	0.426
Another gender	0.008	-
Incomplete postsecondary	0.446	0.437
Complete postsecondary	0.492	0.532
Graduate degree	0.062	0.027
Self employed	0.138	0.167
Employee	0.722	0.672
Not employed	0.140	0.155

Notes: This table compares the characteristics of participants in the experimental sample with those of the relevant population in Argentina's Household Survey (*Encuesta Permanente de Hogares*, EPH). The EPH sample is restricted to individuals aged 25–45 who have completed at least secondary school. To ensure comparability across datasets, education groups are defined using a classification that can be consistently constructed in both samples: low education includes individuals whose highest completed level is secondary school, while high education includes individuals with at least some postsecondary education. EPH statistics are computed using survey weights.

Table A.6: Effect on follow-up scores

	Overall (z-score)	Open ended (z-score)	Multiple choice (z-score)
Low education			
$\overline{Y}_L^C$	0.000	0.000	0.000
	(0.060)	(0.060)	(0.060)
$\overline{Y}_L^T$	0.171***	0.237***	-0.078
	(0.063)	(0.061)	(0.069)
$\overline{Y}_L^T - \overline{Y}_L^C$	0.171**	0.237***	-0.078
	(0.087)	(0.086)	(0.091)
High education			
$\overline{Y}_H^C$	0.300***	0.260***	0.208***
	(0.058)	(0.057)	(0.057)
$\overline{Y}_H^T$	0.371***	0.378***	0.133**
	(0.056)	(0.055)	(0.060)
$\overline{Y}_H^T - \overline{Y}_H^C$	0.071	0.118	-0.075
	(0.080)	(0.079)	(0.083)
High vs. Low education			
$\overline{Y}_H^C - \overline{Y}_L^C$	0.300***	0.260***	0.208**
	(0.083)	(0.083)	(0.083)
$\overline{Y}_H^T - \overline{Y}_L^T$	0.200**	0.141*	0.211**
	(0.084)	(0.082)	(0.091)
$(\overline{Y}_H^T - \overline{Y}_H^C) - (\overline{Y}_L^T - \overline{Y}_L^C)$	-0.100	-0.119	0.003
	(0.118)	(0.116)	(0.124)
Observations	1174	1174	1174

Note: This table reports the results of estimations where the dependent variable is the score in the follow-up questions, standardized relative to the low-education control group: overall follow-up score (column 1), average across ten grading iterations of the open-ended question score (column 2), and multiple-choice score (column 3). The top and central panels show the mean of the control group and the difference between the treatment and control group for the low- and high-education groups, respectively. The bottom panel shows the difference between control and treatment means for the high- and low-education groups, as well as the difference between the treatment effects of both education groups. * significant at 10%; ** significant at 5%; *** significant at 1%.

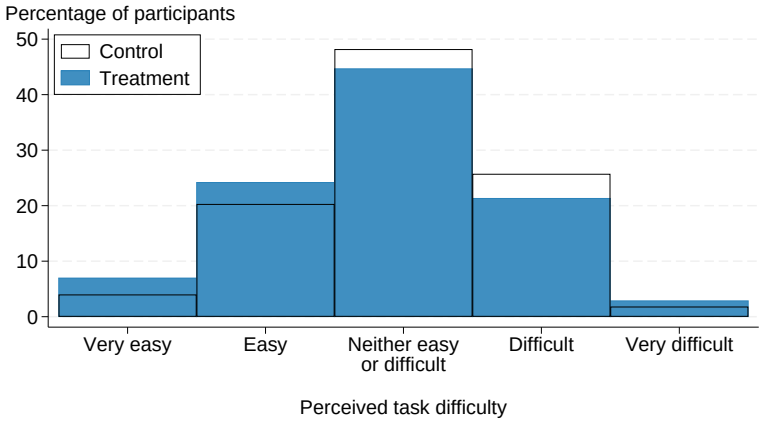
Table A.7: Overall task score and AI use

	(1)	(2)	(3)	(4)	(5)	(6)
Thinking through response						
Share of task components with any AI assistance	1.495***					
	(0.089)					
Detailed instructions for thinking through task (std. index)		0.250***				
		(0.039)				
Generating the output						
Explicitly requested AI to draft answer			0.569***			
			(0.095)			
Requested AI to extend or complete user output			0.328			
			(0.242)			
Requested AI to edit user output			0.586**			
			(0.246)			
Detailed instructions for drafting response (std. index)				0.173***		
				(0.029)		
Use of AI output						
Final output full copy/paste of AI content					1.672***	
					(0.577)	
Final output partial copy/paste of AI content					1.560***	
					(0.579)	
Final output paraphrases AI content					0.923	
					(0.591)	
Workflow and orchestration of conversation						
High initial specificity						0.577***
						(0.078)
Iterative engagement						0.289***
						(0.080)
Structured workflow						0.311***
						(0.095)
Observations	471	471	471	471	471	471
R^2	0.422	0.105	0.082	0.043	0.090	0.151

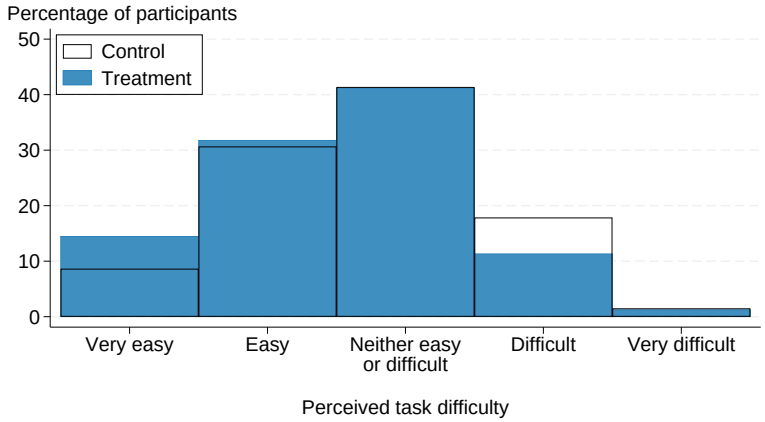
Notes: The dependent variable is the average task score across ten grading iterations, standardized relative to the low-education control group. The sample consists of treated participants who interacted with the assistant. All regressors are constructed from the conversation logs. Drafting-request indicators omit no explicit drafting request; use-of-output indicators omit no direct use of AI content. Indices are standardized relative to the low-education group. Appendix E gives complete definitions. When all AI-use measures are included jointly, the regression has an R^2 of 0.49. * significant at 10%; ** significant at 5%; *** significant at 1%.

Figure A.1: Perceived task difficulty by treatment and education group

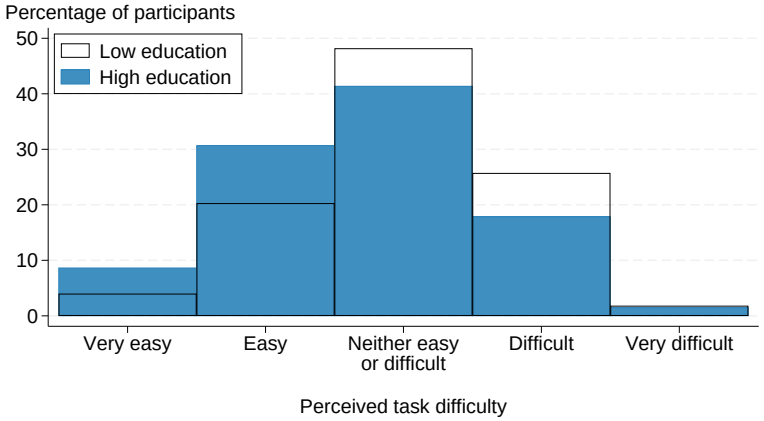
(a) Low-education sample: treatment vs. control



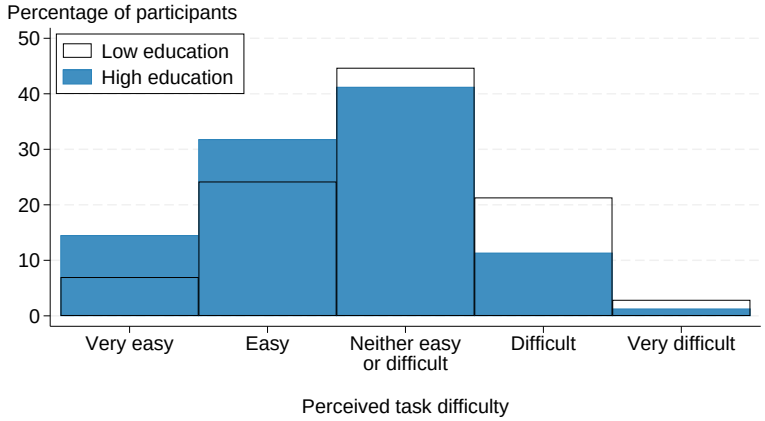
(b) High-education sample: treatment vs. control



(c) Control group: high- vs. low-education

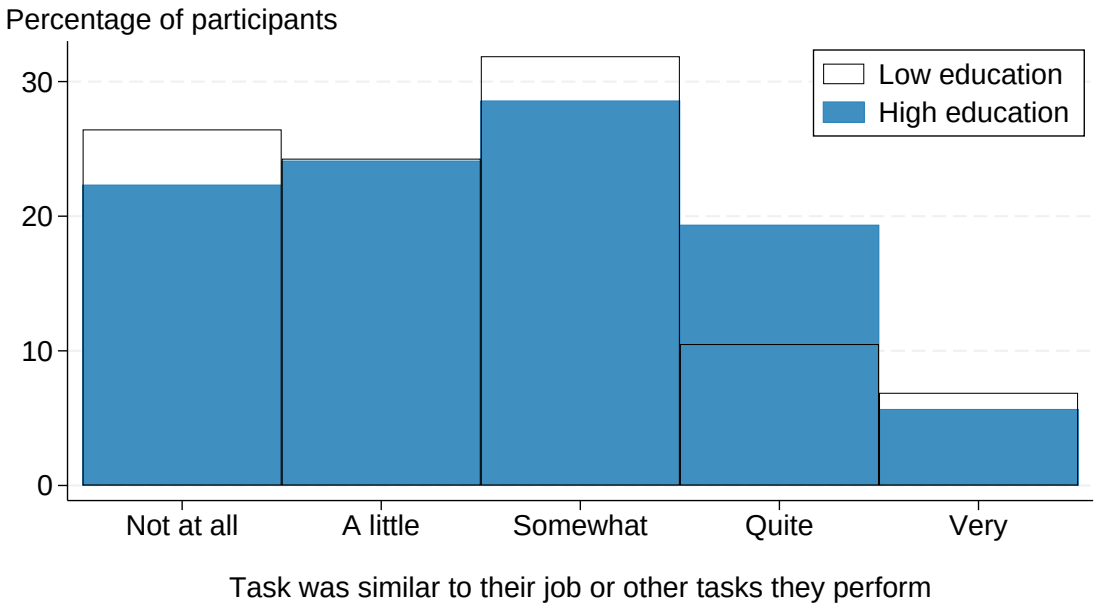


(d) Treatment group: high- vs. low-education



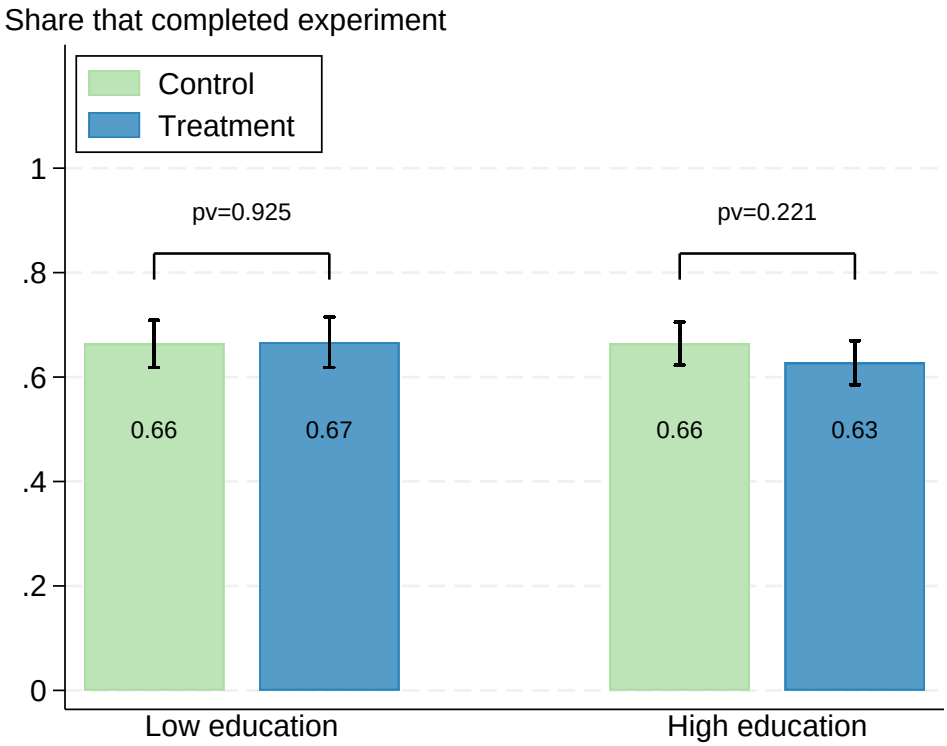
Notes: The top figures show the distribution of the perceived difficulty of the task by treatment group for the sample of low-education (left) and high-education individuals (right). The bottom figures show the analogous distribution by education group for the sample of control group (left) and treatment group individuals (right).

Figure A.2: Familiarity with the task in the control group by education level



Notes: This figure shows the distribution of participants' familiarity with the task by education level for the sample of individuals in the control group.

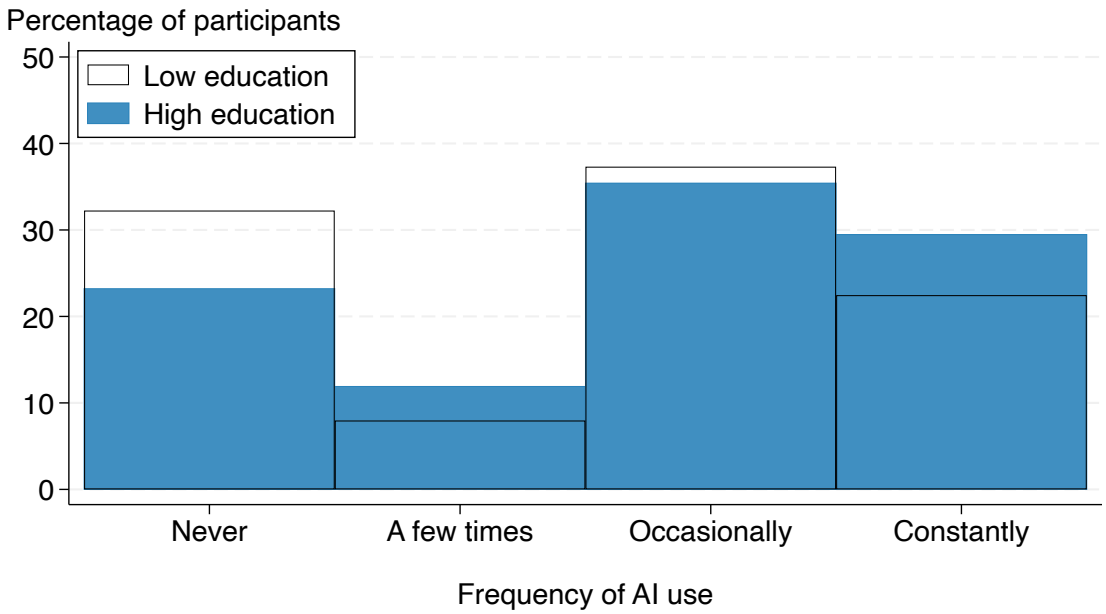
Figure A.3: Experiment completion rate by education and treatment group



F-test of equality of completion rate across groups: p-value=0.530

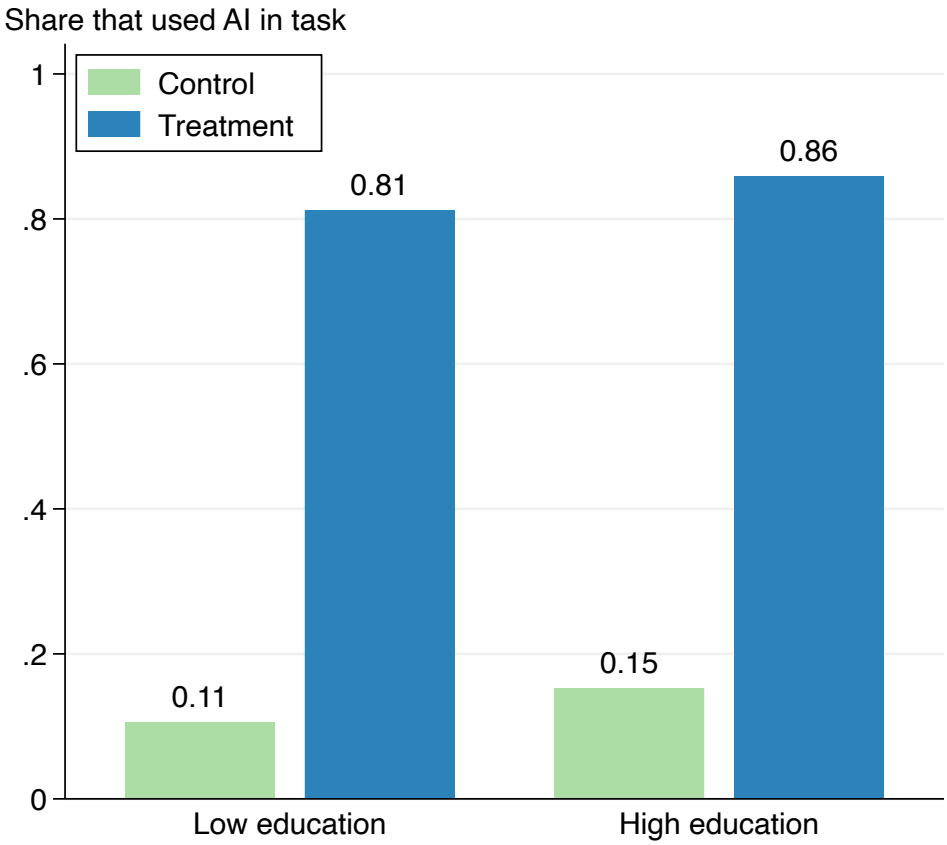
Notes: This figure shows the share of participants that finished the experiment, for each education and treatment group, along with the 95% confidence intervals. Above each education group, we report the p-value for the difference between the treatment and control groups within that education category.

Figure A.4: Frequency of previous AI use in the control group by education level



Notes: This figure plots the frequency of prior AI use for low- and high-education participants in the control group.

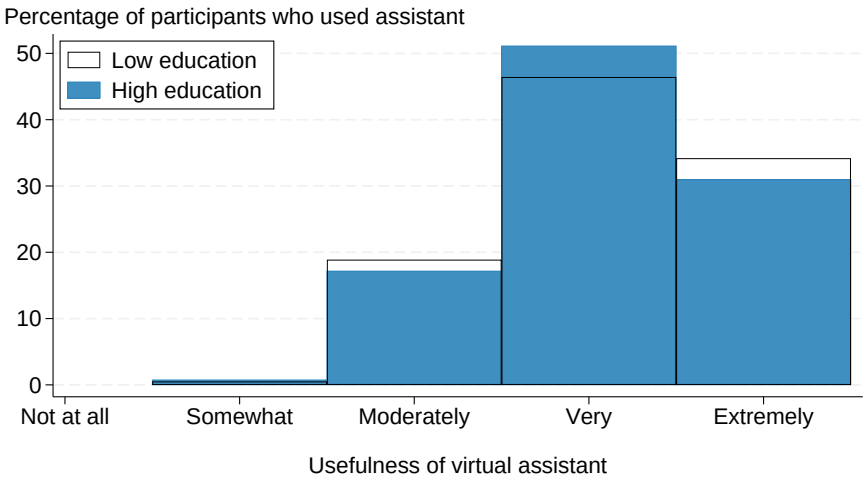
Figure A.5: AI use by education and treatment group



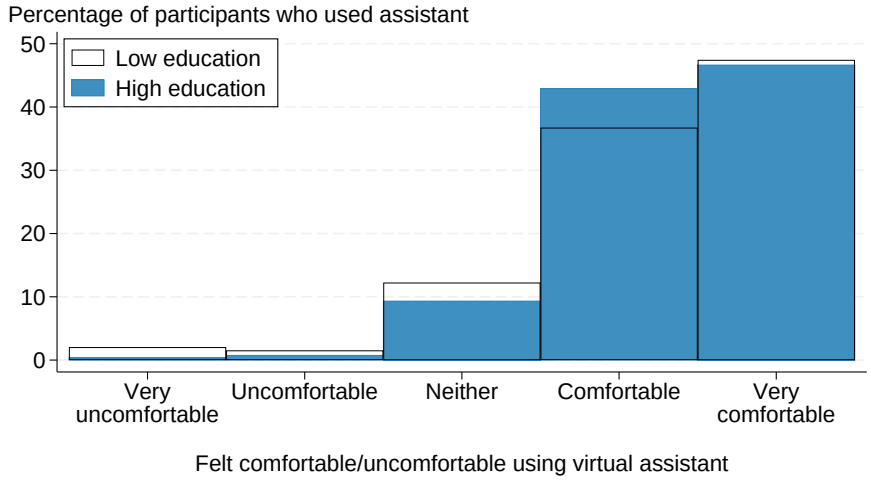
Notes: This figure shows the share of participants that used AI to complete the experimental task, for each education and treatment group. In the treatment group, a respondent is considered to have used AI if they had at least one interaction with the virtual assistant. In the control group, we assume a respondent used AI if they exited the page during the task and there is direct evidence of AI use in their final responses or snapshots of their response, or if they exhibited a maximum word-count change between consecutive snapshots of their task response larger than 75 words (the 25th percentile among treated AI users).

Figure A.6: Self-reported usefulness and comfort of using the AI assistant in the treatment group by education group

(a) Usefulness of AI assistant

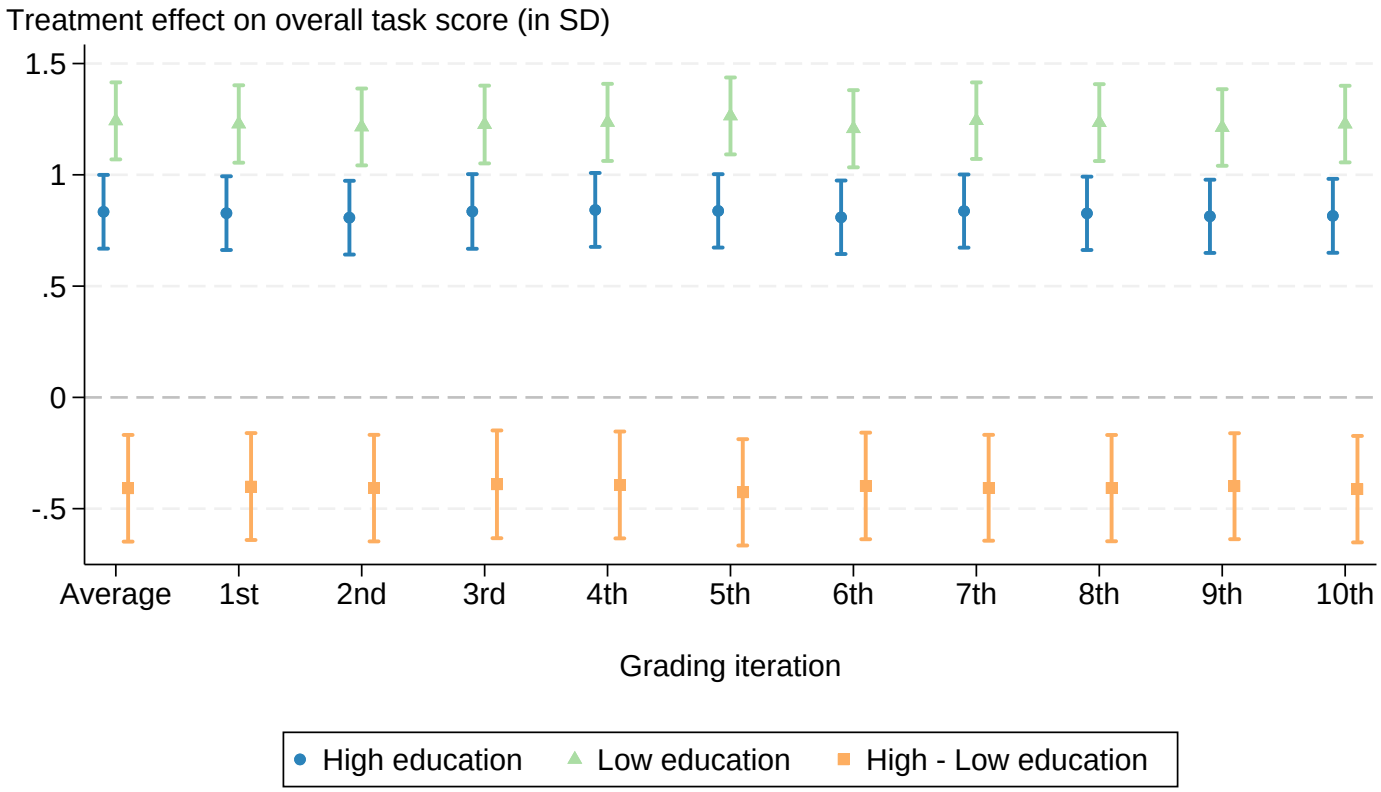


(b) Comfort while using AI assistant



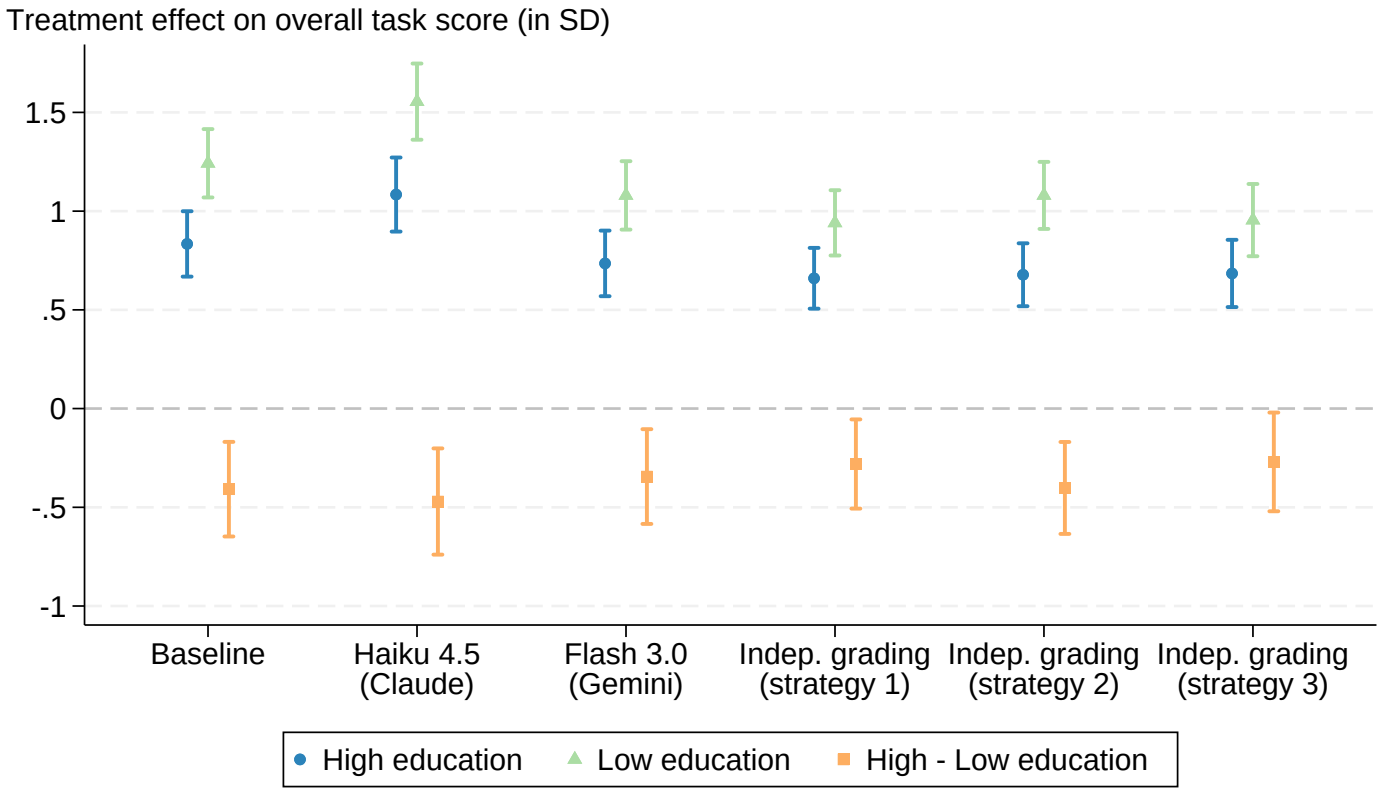
Notes: The top figure shows the distribution of perceived usefulness of the AI assistant for participants in the treatment group that used it during the task, for low- and high-education individuals. The bottom figure shows the distribution of self-reported comfort in using the AI assistant, for the same sample.

Figure A.7: Treatment effects by education group – Sensitivity of results to grading iterations



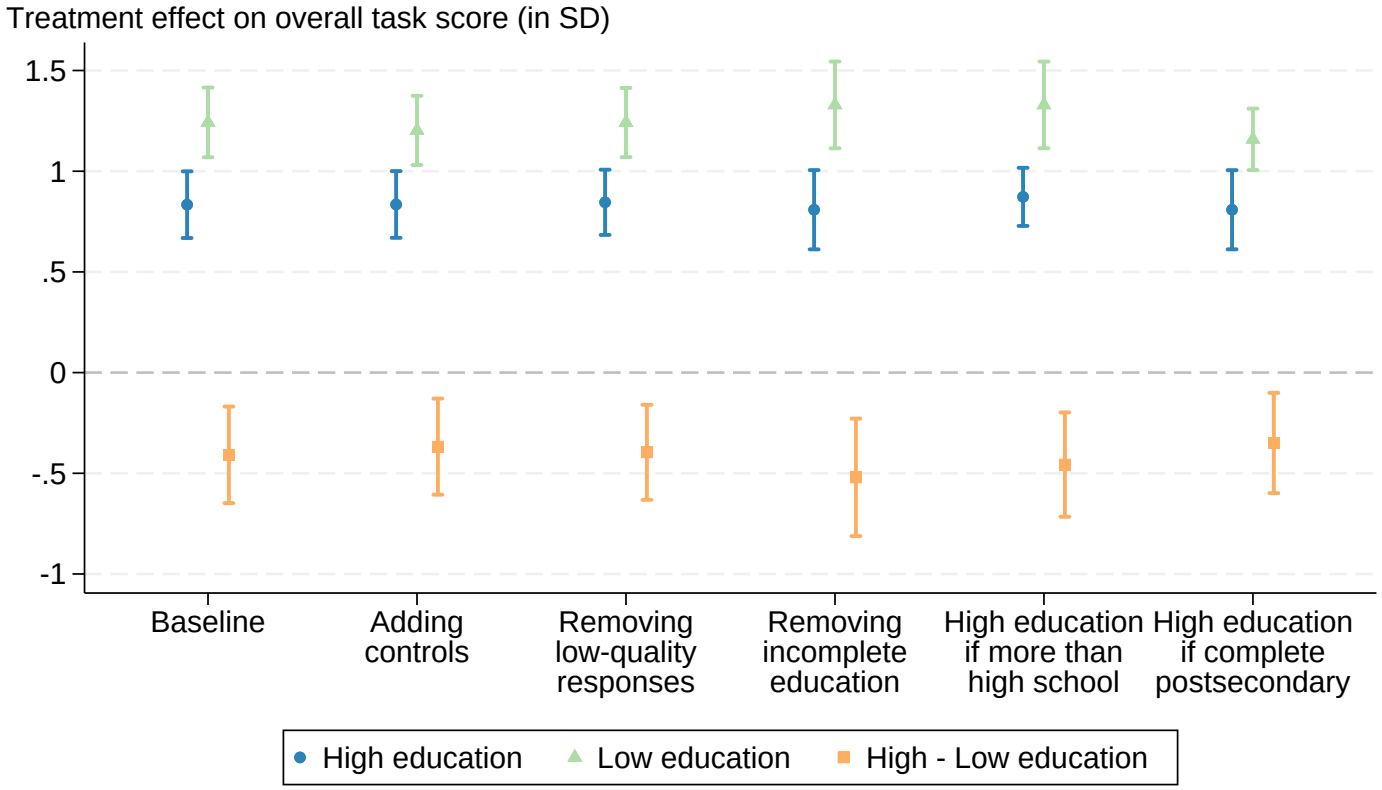
Notes: This figure plots treatment effects on the overall task score (standardized relative to the low-education control group) for low-education participants, high-education participants, and the difference between these effects, estimated separately using each individual grading iteration as the final score. Point estimates are shown with 95% confidence intervals. The first set of estimates corresponds to the average of ten independent grading iterations used in the main analysis. The remaining sets report results for each individual iteration.

Figure A.8: Treatment effects by education group – Robustness to alternative grading procedures



Notes: This figure summarizes a series of robustness checks for the estimated treatment effects on the overall task score (standardized relative to the low-education control group). The figure reports the estimated treatment effects for low-education participants, high-education participants, and the difference between these effects, together with 95% confidence intervals. "Baseline" corresponds to the estimates reported in the main analysis. "Haiku 4.5" and "Flash 3.0" report estimates obtained by grading all responses in a single run using Anthropic's Haiku 4.5 and Google's Gemini-Flash 3.0 models, respectively. "Independent grading" reports the estimates using the three AI-assisted grading strategies developed by an independent researcher not involved in the project.

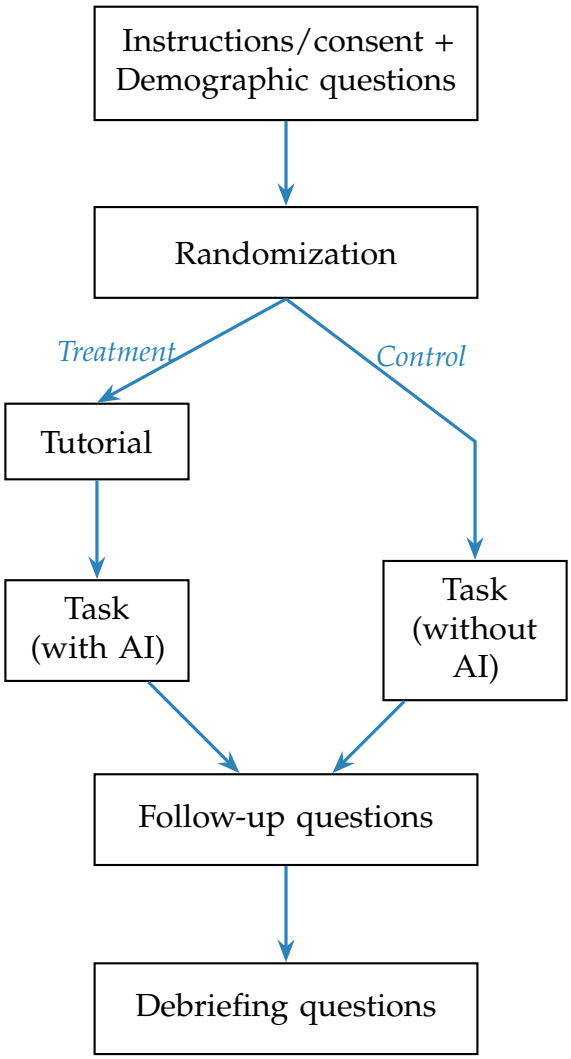
Figure A.9: Treatment effects by education group – Additional robustness checks



Notes: This figure summarizes a series of robustness checks for the estimated treatment effects on the overall task score (standardized relative to the low-education control group). The figure reports the estimated treatment effects for low-education participants, high-education participants, and the difference between these effects, together with 95% confidence intervals. "Baseline" corresponds to the estimates reported in the main analysis. "Adding controls" reports estimates from regressions that include task fixed effects, age, gender, employment status, educational attainment, and work experience. "Removing low-quality responses" excludes from the estimation sample the 25 individuals whose responses are identified as low quality. "Removing incomplete education" excludes participants with incomplete postsecondary education. "High education if more than high school" and "High education if complete postsecondary" present results using alternative definitions of the high-education group.

B Further details of the experiment

B.1 Overview of the experiment



B.2 Sample size and construction of the analysis sample

Table B.1: Sample sizes

Stage		*N*
Recruitment and advancement throughout the experiment	Individuals invited to participate	74,449
	Entered the experimental platform (unique IP)	3,002
	Entered using a computer	2,445
	Consented to participate	1,858
	Completed demographic questions	1,802
	Finished the experiment	1,188
Exclusions	Excluding 1 copied response	1,187
	Excluding 3 submissions when the AI assistant was not functional	1,184
	Excluding 10 unintelligible submissions	1,174
Final sample		1,174

Table B.1 summarizes the flow of participants from recruitment to the final sample. In total, 74,449 individuals were invited to participate in the experiment (pooling recruitment across the three panel survey companies). Invitations relied on pre-screening based on age and educational attainment. Specifically, the panel companies were instructed to recruit individuals aged 25–45 with at least a completed high-school degree.

There were 3,662 entries to the experimental platform, of which 3,002 correspond to unique IP addresses. To avoid potential contamination across observations from the same IP address, we adopt a uniform rule and keep only the first observation for each IP address. In addition, when the same individual appears multiple times in the recruitment records of a survey company (i.e., the same panelist ID) but with different IP addresses, we keep only the first observation. Of these 3,002 unique IP observations, 2,445 entered the experimental platform using a personal computer. Among those, 1,858 individuals consented to participate, and 1,802 completed the demographic questions. A total of 1,188 participants completed the experiment.

We then apply additional data-quality exclusions to arrive at our final experimental sample. We exclude 1 copied response, identified as a submission from an IP address associated with other submissions containing identical answers, and 3 submissions recorded during a short period when the AI assistant was temporarily unavailable.[1] Finally, we classify 10 submissions with unintelligible responses (e.g., "wfwrfrffewqwew" or "yteg3wv") as non-completions. After these restrictions, we arrive at our final sample of 1,174 participants.

Our preregistration specified a conservative target sample size of 600 (300 low-

[1]The AI assistant was hosted on Render. Due to a hosting-related change affecting DNS-based database connections, the assistant could not connect to its database and was unavailable from October 24, 2025 at 12:32:52 to October 27, 2025 at 12:23:26 (Buenos Aires time).

and 300 high-education). However, our main objective of estimating differential effects between the high- and low-education groups is demanding in terms of statistical power. Accordingly, we noted in the preregistration that recruitment might exceed the conservative target if fieldwork proceeded smoothly despite the logistical constraints of requiring participants to perform the task using a personal computer—a high fraction of most panel surveys' subjects respond on mobile devices, which were not suitable for the nature of our tasks. In practice, data collection progressed better than anticipated, resulting in the optimistic scenario of larger samples for both education groups.

B.3 Use of AI in the control group

A potential concern is that participants in the control group used AI assistance. To prevent this, we instructed all participants not to consult external sources or seek help from others at the start of the experiment and immediately before the task. To monitor compliance, we tracked whether they exited the experimental module; those who did saw a popup reminder that leaving the page was not permitted.

Direct evidence of AI use was assessed using both the response snapshots and participants' final responses. Concretely, we classify control-group participants as noncompliant if they exited the task page and displayed direct evidence of AI use in their submission (e.g., emojis, bullet-point formatting, or excerpts of an AI conversation).

To detect AI use that is less apparent, we computed the largest word-count increase between consecutive response snapshots taken every minute for each participant.[2] Importantly, paraphrasing AI output to avoid detection would only evade this detection strategy if participants never paste the AI-generated text into the response box, since we observed minute-by-minute snapshots of responses rather than only the final submission.

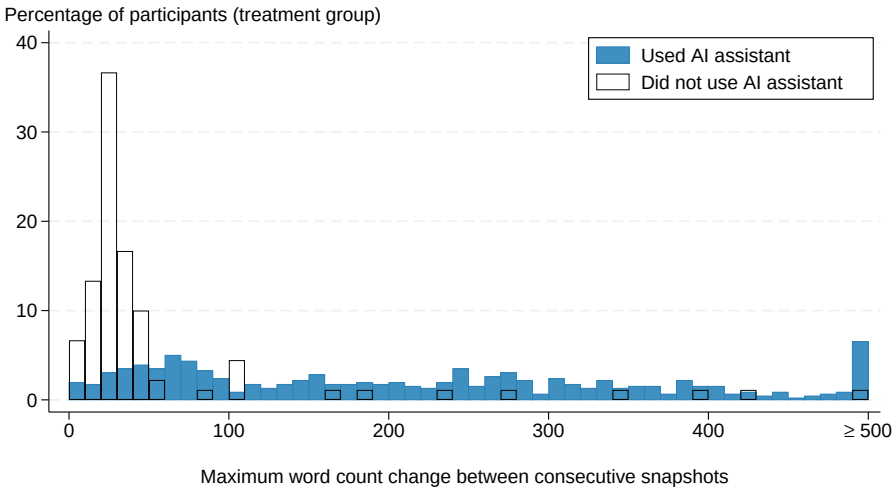
Figure B.1 below shows the distribution of these maximum word-count changes. As expected, treated participants who used the AI assistant exhibit much larger word-count jumps than those who did not (median 198 vs. 28), as seen in the top panel. The bottom panel compares control participants who exited the page during the task (31%) with those who did not (69%). Most control-group participants who exited the page show minor word count changes, similar to those who did not exit the page. However, a small subset of those who exited display sharp word-count increases. We assume that control-group participants sought external AI assistance if they exited the task page and have direct evidence of AI use or a maximum word-count change above 75 words (the 25th percentile among treated AI users).

Using only direct textual evidence, we identify 6% of control participants as noncompliant. Using only the snapshot-based proxy, this share rises to 12%. Combining both sources of evidence yields an overall noncompliance rate of 13% (11% among low-education and 15% among high-education participants).

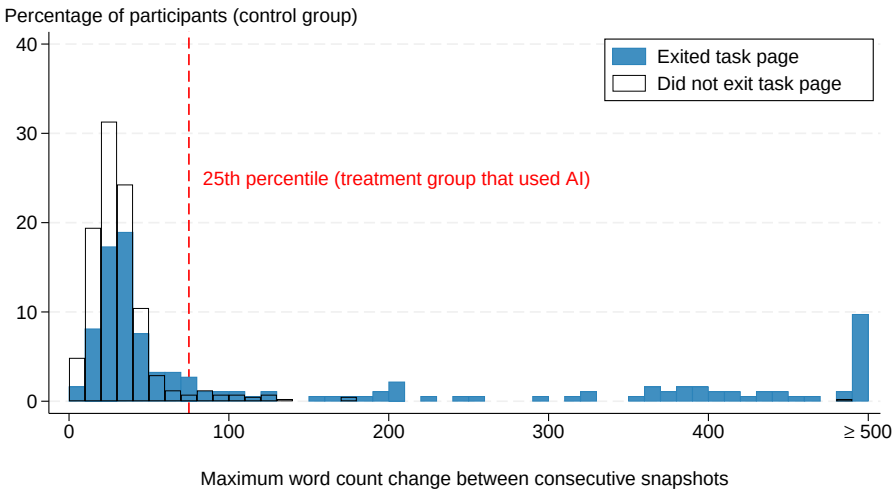
[2] If no snapshot was available (because the participant submitted the task before the first snapshot was taken), we used the word count of the submitted task as the maximum change.

Figure B.1: Comparison of maximum change in word count between consecutive snapshots during task

(a) Used vs. did not use the AI assistant (treatment group)



(b) Exited vs. did not exit the page during the task (control group)



C Further analysis

This section provides additional detail on results summarized briefly in the main text. Section C.1 discusses task-performance subcomponents and related outcomes. Section C.2 describes the results on the subcomponents of the non-AI-assisted follow-up.

C.1 Effects of AI assistance on task-performance subcomponents

Content and writing scores. We find similar effects across both subcomponents of the overall task score. Consistent with the intended difficulty gradient of the task, baseline differences are present in both components. In the control group, high-education participants score 0.503 SD higher in writing and 0.525 SD higher in content, with both gaps statistically significant at the 1% level. AI assistance produces larger gains for low-education participants: their treatment effect exceeds that for high-education participants by 0.382 SD in writing and 0.389 SD in content, with both differences significant at the 1% level. Among treated participants, high-education individuals still outperform low-education individuals by 0.122 SD in writing and 0.136 SD in content, although we are underpowered to detect these differences (p-values are 0.127 and 0.122, respectively). Thus, the point estimates indicate substantial but incomplete convergence in both dimensions. Importantly, the main result is not driven solely by improvements in writing quality. A similar equalizing pattern appears in content as well.

Identification of the root cause of the problem. The preregistered root-cause indicator, a component of the content score, yields the same pattern. The indicator equals one if the participant correctly identifies the root cause of the problem in the assigned task, and zero otherwise. Without AI, 49.5% of low-education participants and 64.7% of high-education participants identify the root cause. With AI, these shares rise to 84.1% and 88.1%, respectively. The treatment effect is 11.2 pp larger for the low-education group and significant at the 5% level. For this measure, the treated-group gap is only 4 pp and statistically insignificant, indicating near-equalization in participants' ability to identify the root cause.

Task completion time. Baseline duration is very similar across education groups: high-education controls finish only 0.274 minutes faster than low-education controls, and the difference is not statistically significant. AI assistance reduces completion time by 0.961 minutes for low-education participants, relative to a control mean of 10.697 minutes, and by 1.514 minutes for high-education participants, relative to a control mean of 10.423 minutes. The reductions are significant at the 10% and 1% levels, respectively, but the difference between the two treatment effects is not statistically significant. As a result, treated high-education participants finish 0.827 minutes faster on average than treated low-education participants, a difference that is marginally insignificant ($p = 0.113$). Task duration is therefore the only reported task outcome that does not display an equalizing pattern; if anything, the point estimates suggest modest divergence in completion speed.

Perceived task difficulty. Perceived difficulty, measured on a five-point Likert scale, is a preregistered secondary outcome and moves in a direction consistent with the

objective performance gains. AI access makes participants in both education groups about 7 pp more likely to report that the task was "very easy" or "easy," with these effects statistically significant at the 10% level and no significant differential effect by education. AI access does not, however, eliminate education-based differences in perceived difficulty, which remain visible among treated participants.

C.2 Impact on non-AI-assisted follow-up subcomponents

All participants completed a set of follow-up questions about the content of the task and their proposed solutions after finishing the experimental task and with no access to the AI assistant. We analyze the treatment effects on the follow-up subcomponents, the open-ended question and the multiple-choice questions. We present the results in Appendix Table A.6.

For the open-ended question, we only find a positive and statistically significant treatment effect among low-education individuals, with a point estimate of 0.237 SD (significant at the 1% level). For high-education individuals, the estimated effect is smaller (0.118 SD) and not statistically different from zero. Although the point estimates suggest larger gains for the low-education group, the estimated difference in treatment effects across education groups of −0.119 is again not statistically significant. These results closely mirror those for the aggregate follow-up score. Furthermore, in this case we also reject equality of open-ended performance between treated high- and low-education participants, as the estimated gap of 0.141 SD is statistically significant at the 10% level.

Finally, for the multiple-choice questions, we do not find evidence of treatment effects in either education group. The estimated effects, equal to −0.078 SD for low-education participants and −0.075 SD for high-education participants, are relatively smaller, statistically insignificant, and nearly identical across groups. As a result, the baseline performance advantage of high-education individuals of 0.208 SD, which is statistically significant at the 5% level, remains largely unchanged after treatment. Among treated participants, the performance gap is 0.211 SD, and it is statistically significant at the 5% level.

Overall, the follow-up results paint a coherent picture. AI assistance generates modest but statistically meaningful improvements in post-task performance only among low-education individuals, with effects concentrated in the open-ended question, which requires participants to articulate the diagnosis they developed during the main task. These gains, however, are not large enough to offset the substantial baseline advantage of high-education individuals.

D AI-assisted grading

In this appendix, we describe the AI-assisted grading procedure used to construct the task and follow-up outcomes analyzed in the paper (Appendix D.1), and analyze the stability of the AI-assisted scores across grading iterations (Appendix D.2). Furthermore, we compare the scores obtained using our AI-assisted procedure with scores obtained by manually applying our grading rubrics to a random sample of responses (Appendix D.3). Finally, we describe alternative AI-assisted grading procedures implemented by an independent researcher, and compare the scores obtained in that procedure with those obtained using our own AI-assisted grading (Appendix D.4).

D.1 Implementation of AI-assisted grading

This appendix describes the AI-assisted grading procedure used to construct the task and follow-up outcomes analyzed in the paper.

D.1.1 Task

We graded participants' responses using an AI-based scoring pipeline powered by OpenAI's GPT-5-mini model. The goal of this system is to implement our preregistered grading scheme in a transparent and reproducible manner, while ensuring that scoring relies exclusively on (i) the task materials and (ii) the participant's response.

During the piloting phase, and strictly following the preregistered scoring structure, we developed task-specific grading rubrics, one for each of the three task versions. Each rubric is stored as a structured JSON object that includes (a) the full task context (email, attached report components, and instructions) and (b) detailed scoring criteria for both the content and writing dimensions. In addition to these criteria, the JSON rubrics include short, illustrative examples for each score level within each sub-item. These examples are designed to leverage the in-context, few-shot learning capabilities of large language models by providing concrete anchors that map rubric language to typical response patterns.[3]

For content, the rubric specifies three diagnostic sub-questions (A1–A3), each scored on a 0–2 scale (maximum 6 points), and a solution component (maximum 4 points). The three diagnostic items correspond to the three diagnosis questions posed in the task email. A1 captures whether the participant directly addresses the first hypothesized mechanism mentioned in the email (the mechanism differs by task version). A2 captures whether the participant addresses the second hypothesized mechanism mentioned in the email (again task-specific). A3 is a general diagnostic question that asks whether the participant identifies other relevant evidence from the attached report beyond the first two hypotheses, for example additional patterns by branch, time, or product segment that help explain what is happening.

The solution rubric explicitly allows participants to propose one or more concrete actions. Each action is evaluated separately under three criteria: whether it addresses the root cause (B1), how specific it is (B2), and whether it is realistic (B3).

[3] "In-context learning" refers to a model's ability to adapt its behavior to a task using instructions and examples provided directly in the prompt at inference time, without updating model parameters. "Few-shot learning" is the special case in which the prompt includes a small number of labeled examples that illustrate the desired input–output mapping.

For writing, the rubric evaluates four components that sum to 10 points: spelling/grammar (0–2), clarity/legibility (0–3), organization (0–4), and register/tone (0–1).

The model is invoked with a fixed system prompt (in Spanish, matching the language of the tasks and answers) that instructs it to behave as an extremely rigorous grader, to avoid inferring missing information, and to return only a valid JSON object in a fixed schema. The system prompt also enforces an "anti-hallucination" rule: if a content element is not present in the participant's response, it must be marked as absent and assigned 0 points. A key feature of the prompt is that, if a response contains multiple proposed actions, the model must (i) detect all the proposed actions and (ii) output B1/B2/B3 scores for each action separately.

In each call, the model receives two inputs: (1) the participant's response and (2) the relevant task-specific rubric, which includes the full task context and the scoring guidelines (including the illustrative examples). The model returns a single JSON object containing (i) diagnostic scores for A1–A3, (ii) an action-level list with one entry per proposed solution with its corresponding B1/B2/B3 scores, and (iii) writing subscores.

Because participants may propose more than one action, we separate two steps. First, the model detects and scores each action. Second, we use the model's action-level outputs to mechanically construct the single solution score used in the analysis. The model is explicitly instructed not to apply cutoff rules and not to select a "best" action. All aggregation is performed by deterministic code outside the model, following the preregistered logic.

Let actions be indexed by $j \in \{1, \ldots, J_i\}$ for participant i. For each action, the model outputs $B1_{ij} \in \{0, 1\}$, $B2_{ij} \in \{0, 1, 2\}$, and $B3_{ij} \in \{0, 1\}$. We compute the participant's solution score $\text{Sol}_i \in \{0, 1, 2, 3, 4\}$ as:

$$\text{Sol}_i = \begin{cases} 0, & \text{if } \max_j B1_{ij} = 0, \\ \max_{j \in \mathcal{J}_i} \left(1 + B2_{ij} + B3_{ij}\right), & \text{otherwise,} \end{cases} \tag{A1}$$

where $\mathcal{J}_i = \{j : B1_{ij} = 1\}$ is the set of actions that address the root cause. This rule implements the preregistered requirement that the solution component receives 0 points if no proposed action addresses the root cause, while allowing participants to receive credit for their best root-cause-addressing action when multiple actions are proposed.

We then compute the content total mechanically as $A1_i + A2_i + A3_i + \text{Sol}_i$ and the writing total mechanically as the sum of the four writing subscores returned by the model. Finally, to prevent very short, incorrect responses from receiving high writing scores, we apply the preregistered deterministic rule that responses with a content score of 0 and length below 200 characters are assigned a writing score of 0. This adjustment is implemented by code outside the model using the model's content and writing outputs.

For transparency and replicability, we provide the full system prompt and the complete task-specific JSON rubrics used for AI-assisted grading in our public GitHub repository at https://github.com/ramirogalvez/ai-skill-gap-experiment-data.

D.1.2 Follow-up question

This subsection describes grading of the open-ended follow-up question. The two multiple-choice follow-up items are graded deterministically using the answer key. We graded the open-ended follow-up question using the same general approach as for

the task, but with a simpler rubric and scoring scale. During the piloting phase, we developed one follow-up grading rubric for each task version. Each rubric is stored as a structured JSON file that contains (i) the follow-up question participants answered, (ii) a benchmark correct answer, and (iii) a three-level rubric defining what constitutes a fully correct answer, a partially correct answer, and an incorrect or irrelevant answer. The rubric also includes short examples for each score level to illustrate how the criteria map into actual responses.

We then asked GPT-5-mini to apply this rubric. We used a fixed system prompt in Spanish that instructs the model to behave as an extremely rigorous grader, to rely only on the rubric and the participant's response, and to avoid inferring information that is not explicitly stated in the response. In each call, the model receives two inputs: (1) the participant's follow-up response and (2) the relevant task-specific follow-up rubric. The model returns a single JSON object containing an integer score in $\{0, 1, 2\}$ and a brief explanation.

In our analysis, we use the model-returned integer score directly as the follow-up open-ended score. Any subsequent transformations, such as weighting, aggregation with the multiple-choice items, or standardization, are computed mechanically in code and are not performed by the model.

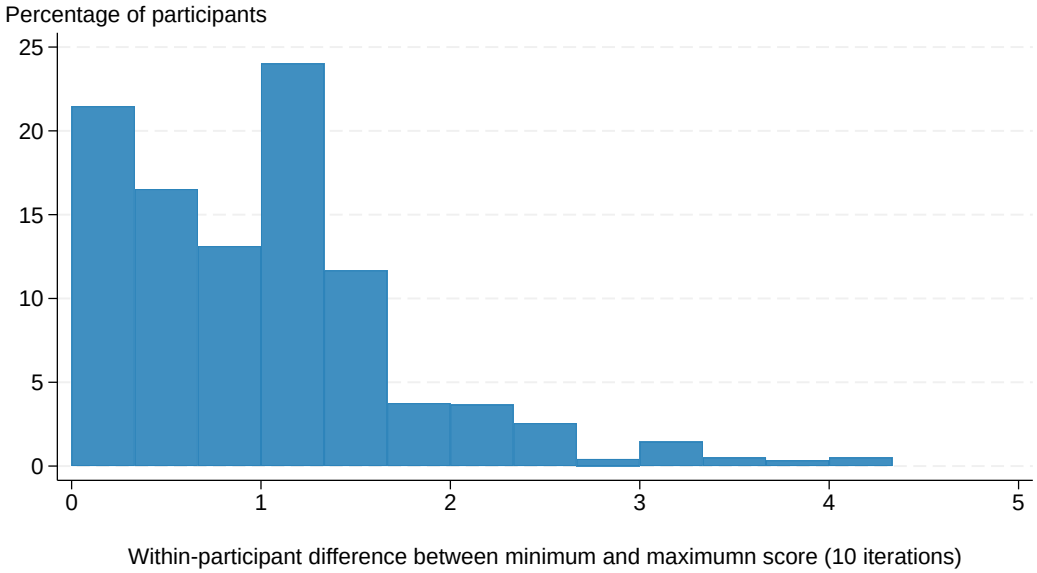
For transparency and replicability, we provide the follow-up system prompt and the task-specific follow-up JSON rubrics in our public GitHub repository at: https://github.com/ramirogalvez/ai-skill-gap-experiment-data

D.2 Variation of scores across grading iterations

As noted in Section 2.3, because AI-generated scores exhibit some variation across runs, we grade each response ten times (independently, that is, with no memory of previous grading runs) and use the average score across the ten iterations as our final measure. This appendix examines the degree of variation in scores across these iterations.

Figure D.1 below shows the distribution of the within-participant difference between the minimum and maximum score obtained across the ten grading iterations. The variation is modest: 83% of responses have less than a 1.5-point difference (out of a maximum of 10 points) between their minimum and maximum scores across iterations. This indicates that while individual grading runs contain some inherent randomness, the scores are generally stable across iterations for most participants.

Figure D.1: Distribution of the variation in scores across AI-grading iterations



Notes: This figure plots the distribution of the difference between the minimum and maximum score for each response across ten AI-grading iterations.

Table D.1 below reports the pairwise correlations between scores across the ten iterations. The correlations are consistently high, ranging from 0.981 to 0.986, indicating strong consistency across different grading runs. Overall, these patterns confirm that averaging across iterations produces a reliable measure of task performance.

Table D.1: Correlation of scores between AI-grading iterations

	Iteration 1	Iteration 2	Iteration 3	Iteration 4	Iteration 5	Iteration 6	Iteration 7	Iteration 8	Iteration 9	Iteration 10
Iteration 1	—	0.984	0.984	0.981	0.984	0.983	0.983	0.983	0.984	0.981
Iteration 2	0.984	—	0.984	0.984	0.981	0.985	0.982	0.984	0.984	0.983
Iteration 3	0.984	0.984	—	0.982	0.985	0.985	0.984	0.985	0.985	0.985
Iteration 4	0.981	0.984	0.982	—	0.983	0.982	0.985	0.984	0.983	0.982
Iteration 5	0.984	0.981	0.985	0.983	—	0.983	0.984	0.984	0.983	0.982
Iteration 6	0.983	0.985	0.985	0.982	0.983	—	0.986	0.984	0.985	0.985
Iteration 7	0.983	0.982	0.984	0.985	0.984	0.986	—	0.986	0.984	0.985
Iteration 8	0.983	0.984	0.985	0.984	0.984	0.984	0.986	—	0.984	0.983
Iteration 9	0.984	0.984	0.985	0.983	0.983	0.985	0.984	0.984	—	0.984
Iteration 10	0.981	0.983	0.985	0.982	0.982	0.985	0.985	0.983	0.984	—

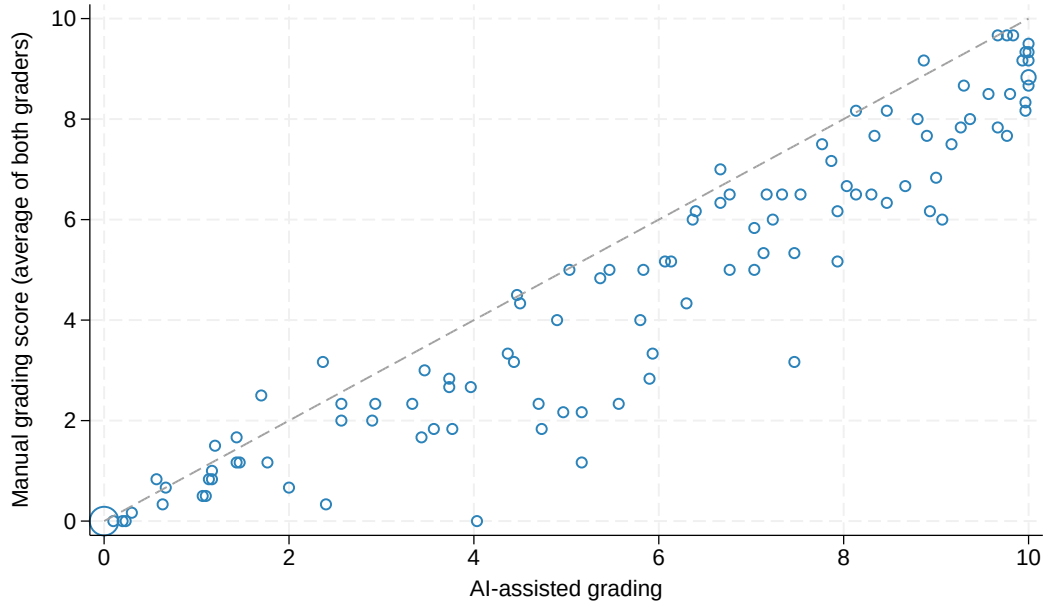
Note: This table reports the correlation across iterations of the AI-generated task scores.

D.3 Comparison of AI-assisted grading and manual grading

To assess the reliability of our AI grading approach, we compare the scores obtained in our AI procedure with human-coded scores for a sample of 117 responses (approximately 10% of our sample). We randomly selected these responses, stratifying by task, education group, treatment group, and whether the grade obtained using AI-grading was above or below the median. The manual grading was independently done by two Economics students (one third year undergraduate and one Master's student) from the Universidad de San Andrés, who used the same task-specific rubrics given to the AI-grader. Importantly, students were only told that participants had to complete a task as part of an experiment, but had no information about the hypothesis of the experiment or the characteristics of the sample.

Figure D.2 below plots the scores given by the human graders (average across both graders) against the scores obtained with our AI-grading procedure (average of 10 iterations). We find a strong alignment between AI-generated and human-coded scores, with most observations clustering close to the 45-degree line.

Figure D.2: Relationship between human-coded scores and AI-generated scores



Notes: This figure plots the human-coded scores (average across both graders) against the AI-generated task scores for a random sample of 117 responses, along with a 45-degree line. The size of each marker is proportional to the number of observations at that coordinate.

Finally, we calculated the Pearson correlation and rank correlation (Spearman's ρ) between the AI-generated scores and the scores given by the human graders. As shown in Table D.2 below, we find a Pearson correlation of 0.951 and a rank-correlation of 0.957 between the AI-generated grades and the average of the manually generated grades, with little variation across graders. Overall, our AI-based grading procedure is reliable and well aligned with human judgment: in the 10% manually graded sample, correlations between our AI-based scores and the two human graders are above 0.9. Section 7 provides the full validation results.

Table D.2: Correlation between human-coded scores and AI-generated scores

	Pearson	Spearman
Average across graders	0.951	0.957
Grader 1	0.918	0.939
Grader 2	0.950	0.951

Note: This table reports the Pearson and Spearman rank correlations between AI-generated task scores and human-coded scores for a random sample of 117 responses. "Average across graders" refers to the average of the two human graders' scores.

D.4 Alternative procedures for AI-assisted grading

To assess whether our findings depend on the specific implementation choices of our AI-assisted grading pipeline, we hired an independent researcher (not involved in this project) to design and run alternative LLM-based grading procedures. The researcher worked from the experiment preregistration (which defines the content and writing components, their weights, and the diagnosis + solution logic) and from the task materials, but did not have access to our detailed correction guidelines. Although we provided the researcher with participants' answers, he was blind to participants' treatment status and education group. These alternative grading procedures were therefore developed independently from ours and provide a robustness check based on different prompts, schemas, and grading workflows.

The independent researcher implemented three strategies, which we describe below.[4] We then compare the scores obtained in our AI-grading procedure to those obtained by the independent researcher.

D.4.1 Strategy 1: LLM-generated rubric and single-step grading

In the first strategy, the researcher used an LLM (GPT-5.1-Pro) to translate the preregistered scoring design and the task instructions into a detailed, LLM-ready grading rubric with explicit anchors and a structured output schema. This translation preserves the preregistered structure but makes it directly usable for large-scale, consistent grading by requiring precise score definitions and standardized outputs. The rubric decomposes performance into content and writing scores. For content, it first scores the three diagnosis questions on a 0/1/2 scale, with criteria specified for each score. Second, it scores the solution component by separately assessing whether the proposed action(s) address the root cause, their specificity, and their realism, and it implements a strict "pass/fail" rule that assigns zero points to the entire solution component if the response does not address the root cause. For writing quality, scores are assigned across the four

[4]We provide the independent researcher's prompts, schemas, and task-specific materials used to implement these strategies in our public GitHub repository at https://github.com/ramirogalvez/ai-skill-gap-experiment-data.

preregistered subdimensions: spelling/grammar, clarity, organization, and tone. The rubric provides specific criteria for each score category, and requires brief justifications for each subscore.

Given this rubric, strategy 1 grades each participant response in a single model call in which the evaluator model must both interpret the task materials (including tables and figures) and apply the rubric to the participant response. The researcher implemented the evaluator using GPT-5.1-thinking-high (verbosity set to high) via the API chat completions interface.

The key strength of this strategy is that it provides a transparent, end-to-end implementation of the preregistered scoring logic with minimal manual steering. The corresponding tradeoff is that it places a relatively high cognitive burden on the evaluator model because it must re-derive the key facts from the task materials for every graded response.

D.4.2 Strategy 2: Same rubric as Strategy 1, plus task-specific "fact sheets" to reduce ambiguity

The second strategy keeps the rubric, schema, score ranges, and aggregation logic exactly the same as in Strategy 1, but changes the information provided to the evaluator model in order to reduce ambiguity and improve stability. In addition to the participant response and the general rubric, the researcher supplied a task-specific fact sheet derived from each task's text, figure, and table (generated with GPT-5.1-Pro). These task-specific materials explicitly list the key facts embedded in the report, including the root cause and other facts relevant for the diagnosis questions, and they highlight common reading errors.

The purpose of this strategy is to avoid requiring the evaluator model to repeatedly rediscover what the chart implies or which numbers matter in the table for each response. Instead, the model can focus on applying the same rubric more consistently, particularly for diagnosis and root-cause identification, where occasional misreadings of the task materials could otherwise add noise. The researcher implemented Strategy 2 using the same evaluator model as Strategy 1 (GPT-5.1-thinking-high, verbosity high). Relative to Strategy 1, the practical advantage of Strategy 2 is improved scoring stability while preserving the same conceptual mapping from responses to points.

D.4.3 Strategy 3: "Grading on a curve" via pairwise comparisons aggregated with Elo (and within-task z-scores)

The third strategy departs from absolute point assignment and instead elicits comparative judgments. An evaluator model is asked to rank groups of responses from the same task version in a strict order (no ties), using the preregistered notion of quality as guidance (content weighted more than writing, diagnosis + solution logic, and the root-cause requirement). The implementation forms groups of 10 responses at a time. From each strict ranking, it constructs a conservative set of pairwise comparisons using only adjacent pairs in the ordering (so a group of size 10 yields 9 binary comparisons). Groups are constructed to balance coverage so that each response participates in approximately 20 pairwise comparisons within its task.

These win/loss outcomes are then aggregated into Elo ratings within each task version. Elo is a standard rating system originally developed for chess that estimates

a latent "skill" (here, response quality) from a sequence of head-to-head outcomes. Intuitively, responses are treated as "players": when response i is ranked above response j, i is recorded as winning a pairwise comparison against j. Elo then updates ratings sequentially after each comparison based on (i) the observed outcome (win/loss) and (ii) the expected probability of winning implied by the current rating difference. Ratings therefore drift upward for responses that consistently win and drift downward for responses that consistently lose. The size of each update is governed by a tuning parameter K, which controls how quickly ratings react to new comparisons.

In the implementation used here, all responses begin with the same initial rating (1000). Expected win probabilities are computed using the standard Elo logistic mapping, and ratings are updated sequentially with a fixed step size (set to 32 by the researcher). Because Elo levels are not directly comparable across separate "tournaments" (here, task versions), the outcome variable is therefore standardized with respect to the low-education control group separately for each task.

The key advantage of Strategy 3 is that it relies on ordinal, comparative evaluation rather than on the absolute calibration of a numeric rubric scale. This can sharpen discrimination among responses that would otherwise bunch together under point-based grading, and it provides a conceptually distinct robustness check in which the primary measurement object is relative quality (and relative distance) rather than exact point attribution by subcomponent.

D.4.4 Comparison of our AI-assisted grading to the independently developed AI-assisted grading

In this section, we compare the scores obtained in our AI grading procedure with the scores obtained in the three strategies developed by the independent researcher. For strategies 1 and 2, we compare the overall score on a 0–10 scale with the average overall score across 10 iterations of our own AI-assisted grading, also on a 0–10 scale. As shown in Figures D.3 and D.4 below, we find a strong alignment between our own AI-generated scores and those obtained by the independent researcher using strategies 1 and 2, with most observations clustering close to the 45-degree line.

Figure D.3: Own and independently developed AI-assisted scores – Strategy 1



Notes: This figure plots the scores obtained by the independent researcher using Strategy 1 against our own AI-generated task scores, all on a 0–10 scale, along with a 45-degree line. The size of each marker is proportional to the number of observations at that coordinate.

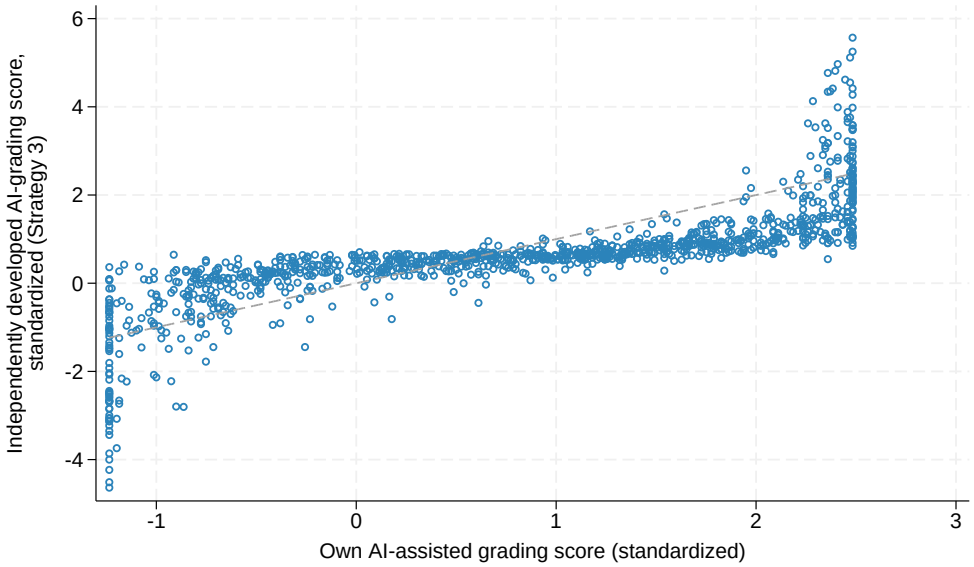
Figure D.4: Own and independently developed AI-assisted scores – Strategy 2



Notes: This figure plots the scores obtained by the independent researcher using Strategy 2 against our own AI-generated task scores, all on a 0–10 scale, along with a 45-degree line. The size of each marker is proportional to the number of observations at that coordinate.

Note that in the case of strategy 3, the final output of the grading procedure is an Elo rating, presented on an arbitrary scale, and not comparable across tasks (as the ranking procedure is conducted within each task). Therefore, to compare the scores under strategy 3 with our own AI-generated score, we first standardize the Elo rating by the mean and standard deviation of the low-education control group (separately within each task), and plot this against our main outcome variable, the standardized overall task score. As shown in Figure D.5 below, the two measures track each other closely, indicating a high degree of agreement across grading approaches. The correspondence is particularly tight over the central part of the distribution. Differences become more visible in the tails, where the Elo-based measure exhibits greater dispersion. This reflects the fact that the Elo rating is not constrained to the 0–10 scale used in our main grading procedure, allowing for more extreme relative distinctions among very low- and very high-quality responses.

Figure D.5: Own and independently developed AI-assisted scores – Strategy 3



Notes: This figure plots the scores obtained by the independent researcher using Strategy 3 (standardized within task by the mean and standard deviation of the low-education control group) against our own AI-generated task scores (standardized by the mean and standard deviation of the low-education control group), along with a 45-degree line. The size of each marker is proportional to the number of observations at that coordinate.

Table D.3 summarizes these comparisons by reporting pairwise correlations across the different grading measures. Consistent with the close visual correspondence shown in Figures D.3 and D.4, the correlations between our AI-generated score and the scores obtained under strategies 1 and 2 are very high, on the order of 0.9 or above, indicating that the alternative rubric-based implementations generate scores that are very similar to those produced by our own AI-assisted grading procedure. For strategy 3, the linear (Pearson) correlation with our own score is lower but still high, at around 0.784, reflecting the unconstrained and relative nature of the Elo measure. At the same time, the rank correlation is very high (0.934), indicating strong agreement in the ordering of responses across the two approaches. Overall, these results indicate that our findings are not sensitive to the specific AI grading implementation used.

Table D.3: Correlation between independently developed and own AI-assisted scores

	Pearson	Spearman
Strategy 1	0.936	0.939
Strategy 2	0.928	0.926
Strategy 3	0.784	0.934

Note: This table reports the Pearson and Spearman rank correlations between our own AI-generated task scores and the scores obtained by the independent researcher under all three strategies. For strategies 1 and 2, we compare the overall score for each strategy on a 0–10 scale with the average overall score across 10 iterations of our own AI-assisted grading, also on a 0–10 scale. For strategy 3, we compare the Elo rating obtained in strategy 3 (standardized within task by the mean and standard deviation of the low-education control group) against our own AI-generated task scores (standardized by the mean and standard deviation of the low-education control group).

E Classification of patterns of AI use

This appendix describes in detail how we classified participants' patterns of interaction with the AI assistant among the 471 treated participants who used the tool during the task. For transparency and replicability, we provide the full system prompt used to classify participants' conversations with the AI assistant in our public GitHub repository at https://github.com/ramirogalvez/ai-skill-gap-experiment-data.

We focus on three dimensions of AI use: how participants relied on the assistant to work through the task, how they used the assistant in generating the final output, and how they orchestrated the interaction with the assistant over the course of the conversation.

Working through the task. We capture how participants use the AI assistant to reason about the task along two main dimensions. First, we measure the extent of assistance requested, defined as the share of task components (diagnostic questions and solution) for which participants asked the assistant for help. We compute this by first coding, for each task component, whether the participant requested AI assistance, and then constructing a weighted average across components, where weights correspond to the share of total points assigned to each component in the scoring rubric.

Second, we measure the quality of prompting in this dimension. We construct a standardized index using inverse covariance weighting following Anderson (2008). This index captures the extent to which participants provide the assistant with detailed instructions that support reasoning through the task. It combines two dimensions of prompting: requests for disciplined reasoning rather than surface-level answers, and the extent to which participants provide contextual information that is not available to the AI assistant but is required to reason through the task. For the first dimension, we take into account whether the participant asks for data-based justifications, requests cross-checking across sources, ensures that all task components are addressed, or prompts the assistant to evaluate multiple solutions, their trade-offs, and their viability in context. For the second dimension, we consider whether the participant provides the assistant with the full email from the boss, the opening paragraph describing the general problem, or the initial description of the assignment. The index is standardized using the mean and standard deviation of the low-education group.

Generating the output. We characterize the role of the AI assistant in the production of the final output along three dimensions. First, we code the type of drafting assistance requested by participants, distinguishing between explicit requests for the assistant to generate the output, requests to extend or complete a user-generated draft, and requests to edit a draft written by the participant. This classification excludes cases in which participants relied on the assistant to think through the response, perhaps using AI-generated text in their answer, but did not explicitly request the assistant to draft any part of their final answer.

Second, we measure the quality of drafting instructions provided to the assistant using an index that captures how tightly participants guide the writing process. This index measures instructions related to output style, including whether participants specify tone, register, length, structure, formatting, or audience, as well as requests that the output sound natural or context-appropriate, as well as constraints imposed on output content, such as requests to preserve key points or evidence, correct mistakes,

prioritize information, or adhere to substantive requirements of the task.

Third, we classify the extent to which AI-generated content is incorporated into the final submission. Final responses are coded according to whether they rely on AI-generated text through full copy-paste—defined as final answers composed entirely of AI-generated text, with no original content added by the participant (from a single or multiple messages)—partial copy-paste, paraphrasing of AI-generated content, or no direct use of AI-generated text.

Workflow orchestration. Beyond what participants ask for and how they use the assistant's output, we characterize how they orchestrate the conversation itself. We develop a classification system based on three dimensions that capture different aspects of conversation management. The first dimension, high initial specificity, is a binary variable that identifies whether the participant's first message provides sufficient context and direction to the assistant. Participants are identified as having high initial specificity if their first message is substantive in length (more than one sentence) and either establishes an explicit frame of reference to the data or presents relevant task context. The second dimension, iterativity, captures whether participants actively build or refine the interaction beyond a single exchange. Participants are identified as having high iterativity if they send more than two messages and, in messages after the first, request that the assistant add elements or refine previous outputs. The third dimension, structuring, measures whether participants deliberately distribute and develop their request across multiple conversational turns. Participants have a highly structured workflow if the conversation includes at least two user messages that substantively develop or elaborate the task—by providing context, posing analytical questions, requesting new outputs, or adding meaningful requirements—rather than merely refining format or confirming earlier responses.

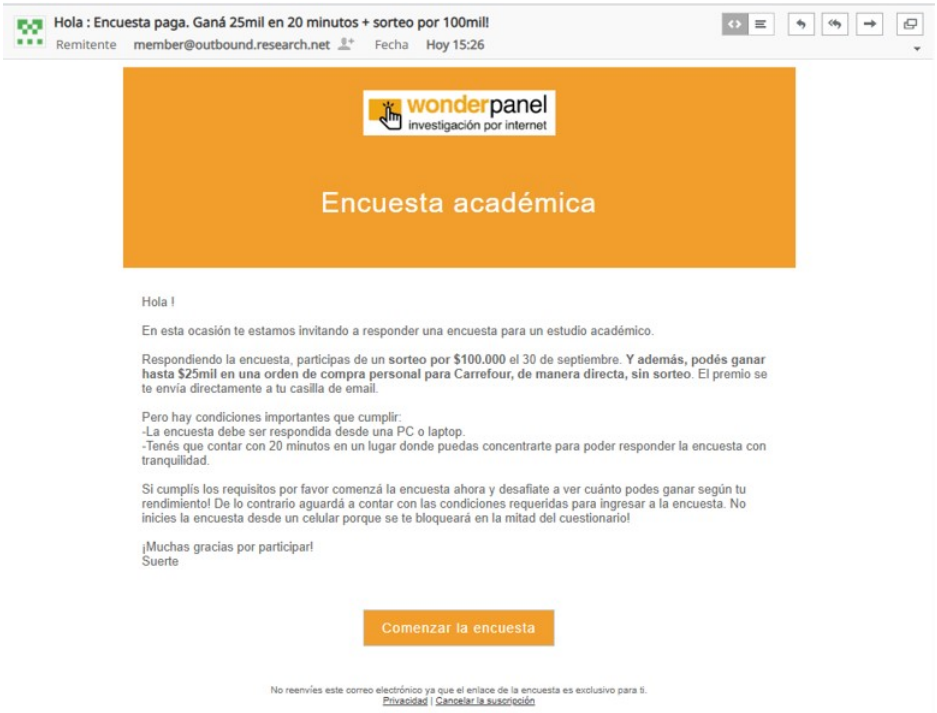
F Supplementary materials

In this section, we present additional details of the experiment. Section F.1 presents screenshots of the recruitment emails sent to participants, and Section F.2 provides English translations of the instructions given to participants. Section F.3 provides an English translation of the tutorial on how to use the AI assistant completed by treated participants, and Sections F.4 to F.6 contain the exact wording of the task and follow-up questions, also translated to English. Finally, Section F.7 presents the system prompt for the AI assistant.

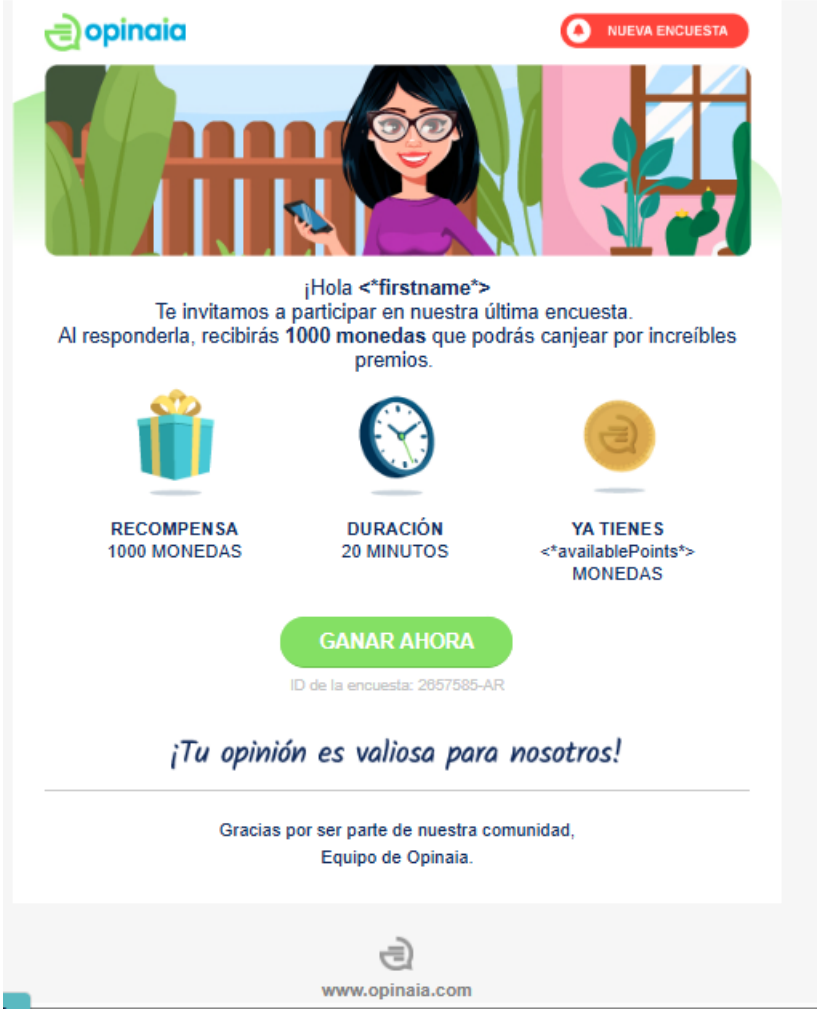
F.1 Recruitment materials

We used three reputable panel survey companies to recruit participants—Wonder, Opinaia, and Offerwise—each of which manages its own survey panel. Each company contacted its panel members to invite them to participate in the experiment. The email described the panel's standard remuneration and stated that participants had to complete the experiment using a computer and that it would take approximately 20 minutes.

Screenshot of the email sent by Wonder:



Screenshot of the email sent by Opinaia:



Screenshot of the email sent by Offerwise:



F.2 Participant instructions

Below, we provide the exact wording (translated from Spanish) of the initial instructions and consent form given to participants at the start of the experiment (Appendix F.2.1), and the instructions given to participants right before the task (Appendix F.2.2) and follow-up questions (Appendix F.2.3).

F.2.1 Initial instructions and consent form

We invite you to participate in this survey, conducted by university professors, to understand how people perform tasks in the workplace.

First, we will ask you some general questions, and then we will ask you to complete a task that simulates typical workplace situations. In addition to the payment for completing the survey, **you can earn up to an additional 25,000 pesos** in Carrefour supermarket gift cards, depending on **how you perform on the task. The better your performance on the task, the higher your reward.** If you do very well, you will earn 25,000 pesos; if you do very poorly, you will receive 0 pesos in additional rewards.

To participate, **we need you to**:

- **use a computer** (it cannot be done on a cellphone),
- **not consult external sources or ask for help**, and
- **complete the survey in one sitting**, without leaving this page (it will take about 20 minutes).

The platform will verify that these conditions are met.

Your responses are confidential and anonymous.

This survey is part of a study approved by the research ethics committees of CEDLAS (Center for Distributive, Labor and Social Studies) and the Nottingham School of Economics. If you have any questions, please contact us at: encuestas@cedlas.org.

Would you like to participate now, and do you meet the conditions? If you click "yes," **you will have one hour to complete the survey** (although **we estimate it will take about 20 minutes**).

F.2.2 Instructions before the task

In the remainder of the survey, we will ask you to complete a task similar to a real work situation and then answer a few brief questions about the task. Please **take the task seriously, as if it were your job**: we will evaluate your responses as a boss would, considering both their content and presentation.

Remember that **you can earn up to 25,000 pesos in Carrefour gift cards depending on your performance on the task and on the subsequent questions**.

You are not allowed to consult external sources or ask for help from other people—we are interested in seeing how you solve the problem on your own. *For treatment group only:* However, **you will have access to a virtual assistant** based on ChatGPT (artificial intelligence), available on the same page as the task. Whether or not you use it is your choice: the reward depends solely on your answers, with or without the assistant.

On the next page, we provide a tutorial on how to use the virtual assistant.

F.2.3 Instructions before the follow-up questions

Next, we will ask you a few short questions about the task you have just completed. **Please remember that you can earn up to 25,000 pesos in Carrefour gift cards, depending on your performance.**

F.3 Tutorial on how to use the AI assistant

The virtual assistant with which you will interact uses artificial intelligence, and knows how to summarize, analyze or edit texts, interpret graphs or tables, and more. To use it, you just have to chat with the assistant as if you were asking someone for help: type your question or request in the chat box and press "Send." You will receive a reply and can continue talking about other aspects, ask for clarifications, etc.

You can use the virtual assistant in different ways to complete the task. For example:

- Share your answer with it, and ask it to edit or simplify it, or to correct it
- Ask it specific questions, or request suggestions on how to solve the task
- Provide it with the entire instructions, and ask it to solve it. Afterwards, you can ask more questions or request clarifications if necessary

You may freely use the assistant's responses —it is there to help you, we will not consider that you are "copying." You may even copy and paste part or all of what it writes.

In the remainder of this page, we present a short practice task (which will not be graded) to guide you on how to use the virtual assistant.

Practice task

This practice task is a simplified version of the task you will do on the next page. It includes an instruction and additional information needed to solve it. Below, you will see the virtual assistant, and finally a response box where you must write your solution to the task. **Important:** the virtual assistant **has access to the additional information, but not to the instructions.**

Instructions: Write a short paragraph with the name of the largest country on the map and the names of the countries that surround it.

Additional information:



Below is the virtual assistant. We want to teach you how to use it. An important part of that is asking the right questions.

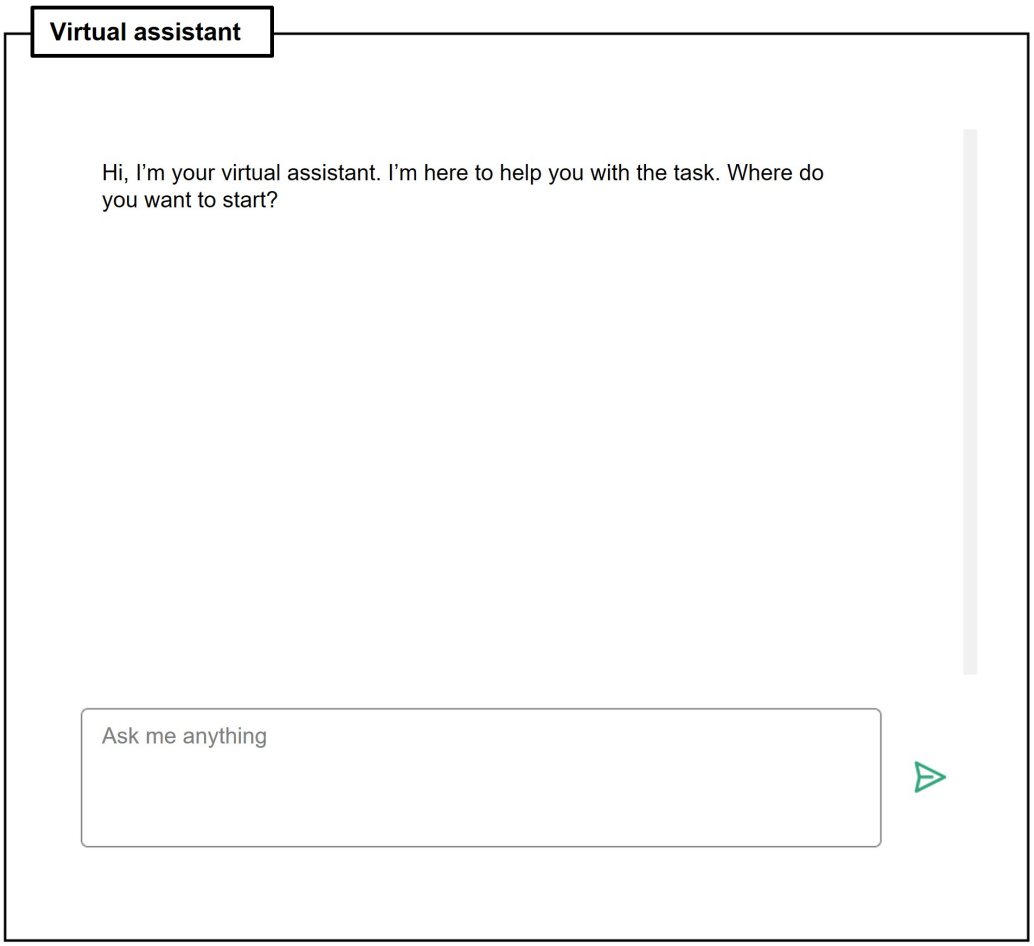
Please **follow the instructions below:**

(1) Type to the virtual assistant: *"Solve the task"*

You will see that the assistant describes the information on the map, but does not give you the answer, because it does not know the instructions.

(2) Now ask: *"What is the name of the largest country on the map, and what countries surround it?"*

(3) Finally, ask: *"Now write this for me in a short paragraph"*



Below is the answer box for the practice task. Write your answer there. If you want, you can directly copy or past the assistant's response, following these steps:

1. Select the text of the assistant that you want to copy (by left-clicking and dragging the mouse over it). Once the text is selected, right-click and choose the "Copy" option.
2. Then, place the mouse over the answer box, right-click again, and select "Paste"

F.4 Task and follow-up questions - Food delivery service

F.4.1 Task

You work in the management of a premium burger delivery restaurant that has two branches (North and South). Each branch has its own employees, who sell burgers exclusively through a delivery platform (similar to PedidosYa). Yesterday you received the following email from your boss:

From: Ramiro González <rgonzalez@gmail.com>

Subject: *problem with online reviews*

Hello, how are you?

I am writing because I have been reviewing the online opinions of our branches, and I am a bit concerned about the North branch: in recent weeks we have been receiving low ratings due to excessive delivery delays. Could you help me understand what is happening?

I attach a report prepared by my assistant to help us better understand the situation. ***Based on the data in this report, I would like your view on the following points:***

- *Could you check whether the problem originates from the delivery platform?*
- *Could it be that the North branch has more orders than the South branch, or that there has been a larger increase in the number of orders in this branch? Or are there no differences in orders between branches?*
- *Is there anything else in the report that you find relevant for understanding what is happening?*

Please substantiate your answers well, since I plan to share this diagnosis with other team members.

Finally, according to your analysis, ***what would you suggest we do to solve this problem?***

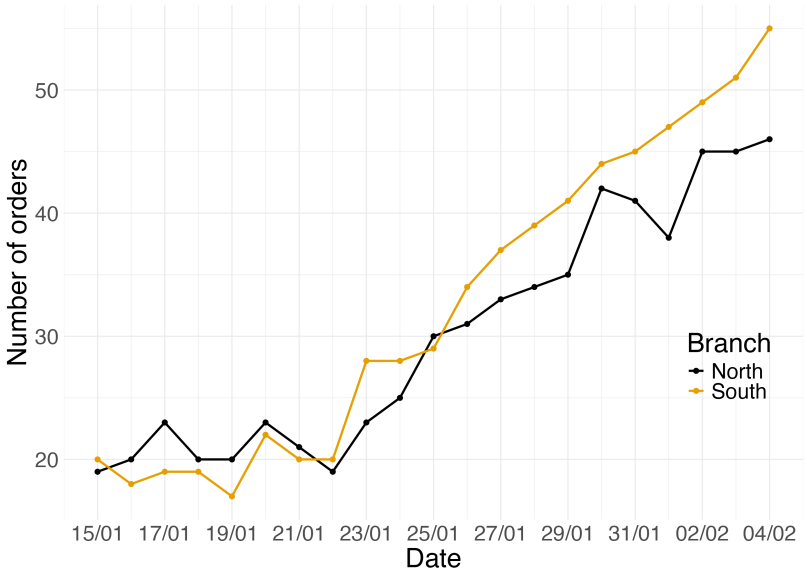
Best regards,

Ramiro

Attached report

Summary of the problem: In recent weeks, the North branch has been receiving low ratings due to excessive delivery delays.

1) Daily number of orders in the last three weeks in the North and South branches:



2) Information on preparation and delivery times for each branch in the last two weeks (provided by the delivery platform):

South Branch

- Average order acceptance time: 9 minutes
- Average preparation time at the restaurant: 18 minutes
- Average delivery time (from handover to the courier until delivery to the customer): 13 minutes

North Branch

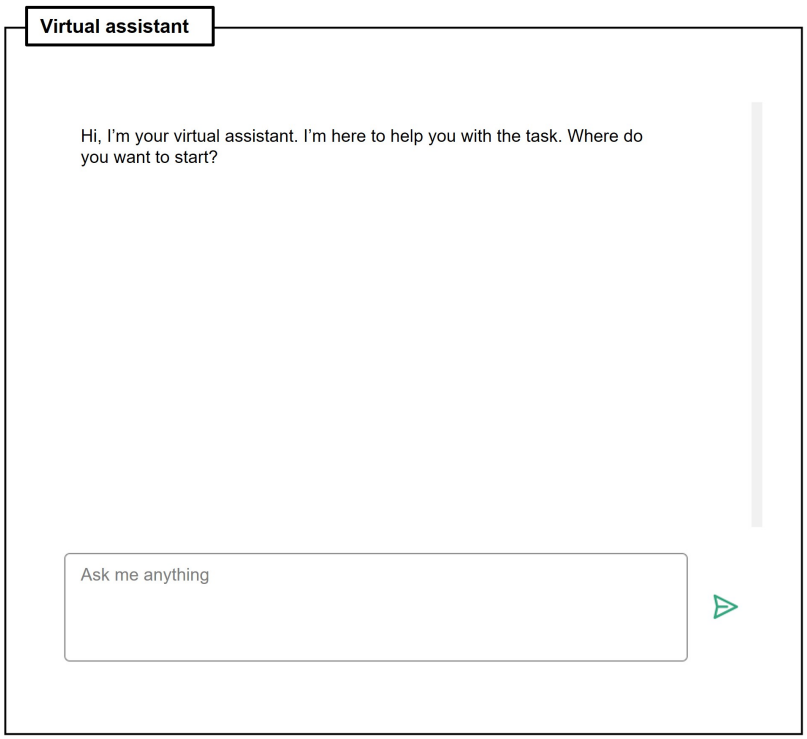
- Average order acceptance time: 10 minutes
- Average preparation time in the restaurant: 29 minutes
- Average delivery time (from handover to the courier until delivery to the customer): 15 minutes

3) Number of employees and their average tenure in each branch:

	North Branch		South Branch	
	Number of Employees	**Average Tenure (months)**	**Number of Employees**	**Average Tenure (months)**
Grill staff	2	0.5	2	4.0
Assembly staff	4	12.0	3	5.0
Cleaning staff	6	15.0	7	7.0
Total	12	11.6	12	6.0

For control group only: **Write an email below with your response to Ramiro.**

For treatment group only: **At the end of the page you will find the space to write your response to Ramiro. Below, you have the virtual assistant, in case you choose to use it. Remember that the virtual assistant only knows the information in the attached report.**



Write an email below with your response to Ramiro.

F.4.2 Follow-up questions

One week after sending your email, you see Ramiro at a meeting. He says:

"How are you? I've been sick these past few days and haven't had the chance to read what you sent me yet. I'm just about to go into a meeting where this topic will be discussed. Could you briefly tell me why the North branch is getting low ratings on the delivery platform?"

Write your response to Ramiro below.

Page break

What did the graph show about the evolution of the number of orders in the two branches?

- Throughout the entire period, the North branch had more orders than the South branch
- In both branches, the number of orders remained constant throughout the entire period
- From a certain point onward, the number of orders increased in both branches
- The number of orders in the South branch remained constant, while those in the North branch increased

Which of the following statements is correct?

- There are more employees in the North branch than in the South branch
- There are fewer employees in the North branch than in the South branch
- There are more grill employees in the North branch than in the South branch
- The number of grill employees is the same in both branches

F.5 Task and follow-up questions - Home appliances store

F.5.1 Task

You work in an Argentina-based online business selling imported household appliances. In January 2025, the owner of the business sends you the following email:

From: Luis González <lgonzalez@gmail.com>

Subject: *problem with sales of portable heaters and fans*

Good morning,

I hope you're doing well. I'm writing because I was reviewing the sales of our products and noticed something unusual with the portable heaters and fans. Historically, our fan sales increase in summer and our heater sales in winter. However, this did not happen in 2024. Could you help me understand what's going on?

I attach a report prepared by my assistant to help us better understand the situation. ***Based on the data in this report, I would like your view on the following points:***

- *To understand whether this problem is specific to these two products, could you check whether in 2024 air conditioner sales were different than usual?*
- *My assistant mentioned that last year there were restrictions on the import of certain electronic products. Could you check whether this might have something to do with it?*
- *Is there anything else in the report that you find relevant for understanding what is happening?*

Please substantiate your answers well, since I plan to share this diagnosis with other team members.

Finally, according to your analysis, ***what would you suggest we do to solve this problem?***

I look forward to your reply.

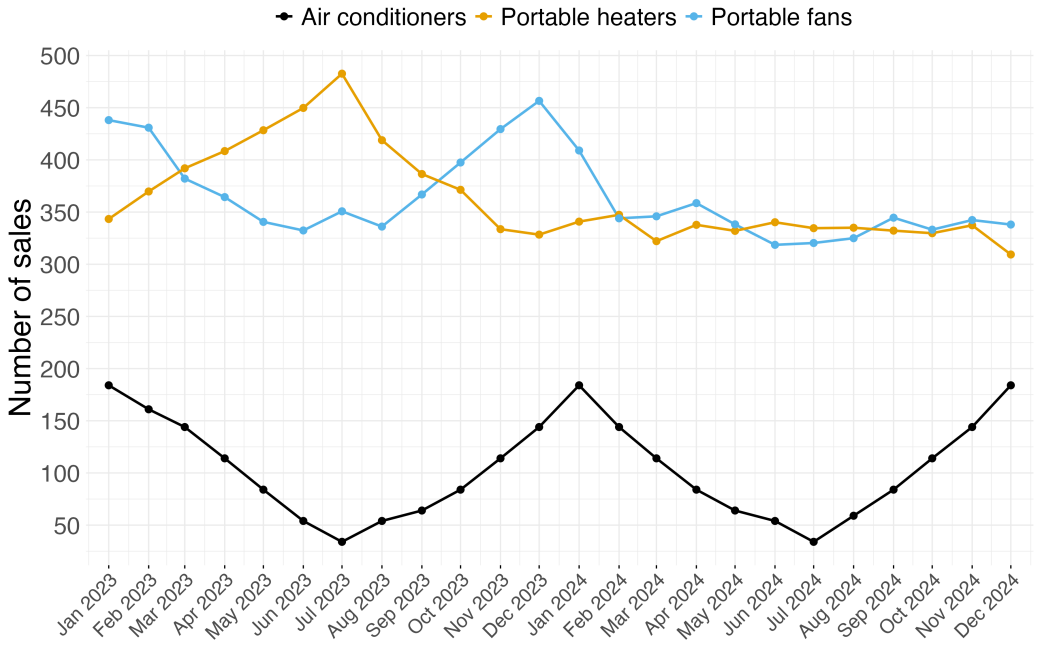
Best regards,

Luis

Attached report

Summary of the problem: Historically, our portable fan sales increase in summer and our portable heater sales in winter. However, this did not happen in 2024.

1) Evolution of sales of air conditioners, portable heaters and portable fans during the last two years:



2) The table below indicates, for each product, the number of units ordered from the importer and delivered by the importer for each semester in 2023 and 2024.

Period	Air conditioner		Portable heaters		Portable fans	
	Units ordered	Units delivered	Units ordered	Units delivered	Units ordered	Units delivered
Jan-Jun 2023	742	741	2,172	2,172	2,145	2,145
Jul-Dec 2023	494	494	2,125	2,127	2,175	2,175
Jan-Jun 2024	687	644	2,182	2,183	2,130	2,131
Jul-Dec 2024	688	619	2,125	2,126	2,150	2,150

3) Each quarter we send an email to our customer base advertising certain products. Below are the products included in this email over the last two years:

- Jan-Mar 2023: Portable fan, television, air conditioner
- Apr-Jun 2023: Refrigerator, washing machine, microwave, portable heater
- Jul-Sep 2023: Portable heater, space heater, dishwasher
- Oct-Dec 2023: Computer, television, air conditioner, portable fan
- Jan-Mar 2024: Electric kettle, air conditioner, portable heater
- Apr-Jun 2024: Television, computer, portable fan, microwave
- Jul-Sep 2024: Dishwasher, washing machine, electric oven
- Oct-Dec 2024: Air conditioner, television, space heater

For control group only: **Write an email below with your response to Luis.**

For treatment group only: **At the end of the page you will find the space to write your response to Luis. Below, you have the virtual assistant, in case you choose to use it. Remember that the virtual assistant only knows the information in the attached report.**



Write an email below with your response to Luis.

F.5.2 Follow-up questions

One week after sending your email, you see Luis at the store. He says:

> *"How are you? I've been sick these past few days and haven't had the chance to read what you sent me yet. I'm just about to go into a meeting where this topic will be discussed. Could you briefly tell me why we are not selling more heaters in winter and portable fans in summer?"*

Write your response to Luis below.

Page break

What did the graph show about the evolution of portable fan sales during 2024?

- They followed the same pattern as air conditioner sales
- They remained constant throughout the year
- They increased at the end of the year
- They declined throughout the year

Which of the following statements is true regarding the email that the company sends to its customer base?

- The products advertised in a period may not match those of the same period in the previous year
- The company sends this email every month
- Portable heaters are never advertised in the email
- Air conditioners are always advertised in the email

F.6 Task and follow-up questions - Cafeteria

F.6.1 Task

You work in the management of "Sabores Express", a business hired to run the cafeteria at the North and South sites of Infotech, a major company. Yesterday you received the following email from your boss:

From: Adriana González <agonzalez@gmail.com>

Subject: *problem with sales at the North site*

Hello, how are you?

I'm writing because I've been looking at the sales from the past few weeks and I can't figure out what's going on in our cafeteria at the North site. Even though Infotech has almost the same number of employees at both sites, we're selling fewer lunches at the North site than at the South. Could you help me understand what's happening?

I attach a report prepared by my team to help us better understand the situation. ***Based on the data in this report, I would like your view on the following points:***

- *Are customers at the North site choosing other lunch alternatives outside the cafeteria to a greater extent?*
- *Have sales at the North site decreased in general or only at a specific moment?*
- *Is there anything else in the report that you find relevant for understanding what is happening?*

Please substantiate your answers well, since I plan to share this diagnosis with other team members.

Finally, according to your analysis, ***what would you suggest we do to solve this problem?***

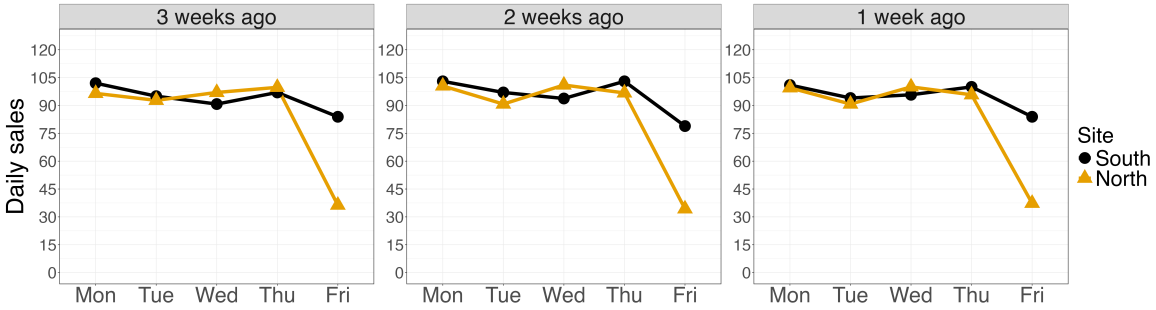
Best regards,

Adriana

Attached report

Summary of the problem: Although Infotech has almost the same number of employees at both sites, fewer lunches are sold at the North than at the South site.

1) Daily number of lunches sold in the cafeteria at each site during the last three weeks:



2) Last week, a survey was conducted among Infotech employees present at the office. They were asked where they got their lunch that day. The results are as follows:

Site	Monday	Tuesday	Wednesday	Thursday	Friday
South	Cafeteria: 80% Other place: 20%	Cafeteria: 76% Other place: 24%	Cafeteria: 75% Other place: 25%	Cafeteria: 80% Other place: 20%	Cafeteria: 78% Other place: 22%
North	Cafeteria: 78% Other place: 22%	Cafeteria: 75% Other place: 25%	Cafeteria: 80% Other place: 20%	Cafeteria: 79% Other place: 21%	Cafeteria: 78% Other place: 22%

3) Infotech has employees in two different areas: administration and sales. Administrative staff can work from home on Fridays, while sales staff are required to be on site from Monday to Friday. The table below shows the characteristics of employees at each site according to their area of work:

	South site		**North site**	
Area	**Number of Employees**	**Average Tenure (years)**	**Number of Employees**	**Average Tenure (years)**
Administration	20	2.0	81	3.0
Sales	105	3.0	44	2.0
Total	**125**	**2.8**	**125**	**2.6**

For control group only: **Write an email below with your response to Adriana.**

For treatment group only: **At the end of the page you will find the space to write your response to Adriana. Below, you have the virtual assistant, in case you choose to use it. Remember that the virtual assistant only knows the information in the attached report.**



Write an email below with your response to Adriana.

F.6.2 Follow-up questions

One week after sending your email, you see Adriana at a meeting. She says:

> *"How are you? I've been sick these past few days and haven't had the chance to read what you sent me yet. I'm just about to go into a meeting where this topic will be discussed. Could you briefly tell me why we are selling fewer lunches at the North site?"*

Write your response to Adriana below.

Page break

What did the graph show about daily lunch sales at both sites?

- Sales at both sites increase on Tuesdays compared to Mondays
- At both sites, Monday sales are higher than the previous week's Friday sales
- On Fridays, sales drop only at the North site
- Every day, the North site sells fewer lunches than the South site

Which of the following statements about Infotech employees is true?

- Only at the North site are Administration staff allowed to work remotely one day per week
- Infotech has Administration and Cleaning staff
- Both sites have the same total number of employees
- There are more employees at the South site than at the North

F.7 Prompt given to the AI assistant

This appendix documents the exact prompts provided to the AI assistant during the experiment. It includes the task-specific prompts for each variation of the experimental task, as well as the prompt used during the tutorial to familiarize treated participants with the assistant's functionality.

F.7.1 Food delivery service task

You are a virtual assistant integrated into a website where people in Argentina must solve a task based on a given problem. As part of this task, users receive a report with the necessary information to solve the problem. Below you can find this report. Your job is to assist users in interpreting this information. Answer all user questions.

The first thing you will receive in the conversation is an image. Treat this image as part of the attached report and not as a message from the user. In other words, treat it the same way as you would treat tables or other information, naturally, and without mentioning that the image was sent as a message. Use it to complement your answers.

Provide clear and helpful responses.

Report

Summary of the problem: In recent weeks, the North branch has been receiving low ratings due to excessive delivery delays.

1) Here goes the image you will receive in the next message. Remember that this message is not from the user, but rather complements the information provided here. The chart title is *"Daily number of orders in the last three weeks in the North and South branches"*.

2) Information on preparation and delivery times for each branch in the last two weeks (provided by the delivery platform):

South Branch

- Average order acceptance time: 9 minutes
- Average preparation time at the restaurant: 18 minutes
- Average delivery time (from handover to courier until delivery to customer): 13 minutes

North Branch

- Average order acceptance time: 10 minutes
- Average preparation time in the restaurant: 29 minutes
- Average delivery time (from handover to the courier until delivery to the customer): 15 minutes

3) Number of employees and their average tenure in each branch:

	North Branch		South Branch	
Category	**No. of employees**	**Average tenure (months)**	**No. of employees**	**Average tenure (months)**
Grill staff	2	0.5	2	4
Assembly staff	4	12	3	5
Cleaning staff	6	15	7	7
Total	12	11.6	12	6

Assist users with their requests, but if a user sends a message that makes no sense (such as "." or "bla bla" or "No" or "3"), do not provide the full solution to the problem. In that case, you can reply with something like: *"How can I help you?"*.

F.7.2 Home appliances store task

You are a virtual assistant integrated into a website where people in Argentina must solve a task based on a given problem. As part of this task, users receive a report with the information needed to solve the problem. Below you can find this report. Your job is to assist users in interpreting this information. Answer all user questions.

The first thing you will receive in the conversation is an image. Treat this image as part of the attached report and not as a message from the user. In other words, treat it in the same way as you would treat tables or any other piece of information, naturally, and without mentioning that the image was sent as a message. Use it to complement your answers.

Provide clear and helpful responses.

Report

Summary of the problem: Historically, our portable fan sales increase in summer and our portable heater sales in winter. However, this did not happen in 2024.

1) Here goes the image you will receive in the next message. Remember that this message is not from the user, but rather complements the information provided here. The chart title is *"Evolution of sales of air conditioners, portable heaters and portable fans during the last two years:"*.

2) The table below indicates, for each product, the number of units ordered from the importer and the number delivered by the importer for each semester in 2023 and 2024:

Period	**Air conditioner**		**Portable heaters**		**Portable fans**	
	Units ordered	**Units delivered**	**Units ordered**	**Units delivered**	**Units ordered**	**Units delivered**
Jan – Jun 2023	742	741	2172	2172	2145	2145
Jul – Dec 2023	494	494	2125	2127	2175	2175
Jan – Jun 2024	687	644	2182	2183	2130	2131
Jul – Dec 2024	688	619	2125	2126	2150	2150

3) Each quarter we send an email to our customer base advertising certain products. Below are the products included in this email over the last two years:

- Jan-Mar 2023: Portable fan, television, air conditioner
- Apr-Jun 2023: Refrigerator, washing machine, microwave, portable heater

- Jul-Sep 2023: Portable heater, space heater, dishwasher
- Oct-Dec 2023: Computer, television, air conditioner, portable fan
- Jan-Mar 2024: Electric kettle, air conditioner, portable heater
- Apr-Jun 2024: Television, computer, portable fan, microwave
- Jul-Sep 2024: Dishwasher, washing machine, electric oven
- Oct-Dec 2024: Air conditioner, television, space heater

Assist users with their requests, but if a user sends a message that makes no sense (such as "." or "bla bla" or "No" or "3"), do not provide the full solution to the problem. In that case, you can reply with something like: *"How can I help you?"*.

F.7.3 Cafeteria task

You are a virtual assistant integrated into a website where people in Argentina must solve a task based on a given problem. As part of this task, users receive a report with the information needed to solve the problem. Below you can find this report. Your job is to assist users in interpreting this information. Answer all user questions.

The first thing you will receive in the conversation is an image. Treat this image as part of the attached report and not as a message from the user. In other words, treat it in the same way as you would treat tables or any other piece of information, naturally, and without mentioning that the image was sent as a message. Use it to complement your answers.

Provide clear and helpful responses.

Report

Problem summary: Although Infotech has almost the same number of employees at both sites, fewer lunches are sold at the North branch than at the South branch.

1) Here goes the image you will receive in the next message. Remember that this message is not from the user, but rather complements the information provided here. The chart title is *"Daily number of lunches sold in the cafeteria at each site during the last three weeks"*.

2) Last week, a survey was conducted among Infotech employees present at the office. They were asked where they got their lunch that day. The results are as follows:

Site	Monday	Tuesday	Wednesday	Thursday	Friday
South	Cafeteria: 80% Other: 20%	Cafeteria: 76% Other: 24%	Cafeteria: 75% Other: 25%	Cafeteria: 80% Other: 20%	Cafeteria: 78% Other: 22%
North	Cafeteria: 78% Other: 22%	Cafeteria: 75% Other: 25%	Cafeteria: 80% Other: 20%	Cafeteria: 79% Other: 21%	Cafeteria: 78% Other: 22%

3) Infotech has employees in two different areas: administration and sales. Administration staff can work from home on Fridays, while sales staff are required to be on site from Monday to Friday. The table below shows the characteristics of employees at each site according to their area of work:

Assist users with their requests, but if a user sends a message that makes no sense (such

Work area	South Branch		North Branch	
	No. of employees	**Average tenure (years)**	**No. of employees**	**Average tenure (years)**
Administration	20	2	81	3
Sales	105	3	44	2
Total	**125**	**2.8**	**125**	**2.6**

as "." or "bla bla" or "No" or "3"), do not provide the full solution to the problem. In that case, you can reply with something like: *How can I help you?"*.

F.7.4 Tutorial

You are a virtual assistant integrated into a website where people in Argentina must solve a task based on a given problem. As part of this task, users receive a report with the information needed to solve the problem. Below you can find this report. Your job is to assist users in interpreting this information. Answer all user questions.

The first thing you will receive in the conversation is an image. Treat this image as part of the attached report and not as a message from the user. In other words, treat it in the same way as you would treat tables or any other piece of information, naturally, and without mentioning that the image was sent as a message. Use it to complement your answers.

Provide clear and helpful responses.

Report

1) Here goes the image you will receive in the next message. Remember that this message is not from the user, but rather complements the information provided here. The chart title is *"Political map of South America"*.

Assist users with their requests, but if a user sends a message that makes no sense (such as "." or "bla bla" or "No" or "3"), do not provide the full solution to the problem. In that case, you can reply with something like: *"How can I help you?"*.